

June 11, 2026 09:00 PM GMT

Greater China Technology Hardware | Asia Pacific

Optics Drives the Board: PCB Beneficiaries of the AI Interconnect Build-out

Although incumbent suppliers, including Unimicron and Shennan Circuits, will likely benefit from the growth in transceiver PCB demand and content expansion, we view ZDT as the most compelling beneficiary, given its potential to capture meaningful share as the industry transitions to higher speeds.

The AI investment cycle is entering a new phase in which optical connectivity is becoming as critical as compute. As hyperscalers scale AI clusters from tens of thousands to hundreds of thousands of GPUs, the number of optical transceivers required to connect these systems is growing rapidly. While investors have largely focused on beneficiaries such as GPUs, networking chips, and optical module vendors, we believe there are also opportunities further down the supply chain: *printed circuit board (PCB) manufacturers*. These companies provide a differentiated way to participate in the secular growth of AI infrastructure spend.

The opportunity is not solely driven by unit growth; content is also rising. As the industry migrates from 400G to 800G and eventually 1.6T optical transceivers, PCB complexity is increasing materially. Higher layer counts (10L to 16L), higher grade CCL (M6 to M8), and the adoption of advanced manufacturing technologies (HDI to mSAP) are driving a step-function increase in PCB content per module (US\$10 to US\$25). As a result, we expect the optical transceiver PCB market to grow substantially faster (83% CAGR) than underlying transceiver shipments (60% CAGR) from CY25-28. We are forecasting AI transceiver PCB TAM growing from ~US\$620M in CY25 to ~US\$3,773M by CY28, and this content-driven growth should also translate into improved profitability and expanding returns for suppliers with the requisite technological capabilities.

ZDT is best positioned to capture share: We expect incumbent leaders to benefit from both volume and content growth; however, we believe the most attractive investment opportunity is Zhen Ding (ZDT). The transition to 1.6T and beyond is raising qualification barriers and increasing the importance of advanced manufacturing know-how, particularly in mSAP processes. As the market expands and customers seek additional qualified suppliers, companies with proven high-end PCB capabilities are positioned to capture disproportionate growth. In our view, this combination of secular AI demand, rising content intensity, and market share gains creates a compelling angle for ZDT, hence we raise our PT to NT\$666, implying 20x CY28 P/E (0.5x PEG). We reiterate our OW on Unimicron, with a raised PT of NT\$1,285, implying 25x CY28 P/E (0.25x PEG), but stay EW on Shennan with a raised PT of Rmb400, implying 25x CY28 P/E (0.6x PEG). We prefer ZDT (15x CY28 P/E) and Unimicron (17x CY28 P/E) but are EW Shennan (25x CY28 P/E) on valuation.

MORGAN STANLEY TAIWAN LIMITED+

Howard Kao

Equity Analyst

Howard.Kao@morganstanley.com

+886 2 2730-2989

Irene Yen

Research Associate

Irene.Yen@morganstanley.com

+886 2 2730-2869

Sharon Shih

Equity Analyst

Sharon.Shih@morganstanley.com

+886 2 2730-2865



GREATER CHINA TECHNOLOGY HARDWARE

Asia Pacific

Industry View

In-Line

Exhibit 1 : What's changed?

Company	Ticker	Rating	PT		Last Price	Upside (%)	EPS change (%)				PEG		
			Old	New	Old		24e	27e	27e	27e	27e	27e	27e

Unimicron	3037.TW	OW	OW	1,285.0	1,285.0	944.0	40%	2%	2%	4%	0.53	0.45	0.25
Zhen Ding	4008.TW	OW	OW	660.0	570.0	522.0	31%	5%	15%	17%	0.42	0.40	0.52
Shennan Circuits	0020.HK	EW	EW	400.0	320.0	383.1	4%	0%	17%	24%	1.21	0.63	0.63

Source: Bloomberg, Morgan Stanley Research estimates. Pricing as of the close on June 10, 2026.

Morgan Stanley does and seeks to do business with companies covered in Morgan Stanley Research. As a result, investors should be aware that the firm may have a conflict of interest that could affect the objectivity of Morgan Stanley Research. Investors should consider Morgan Stanley Research as only a single factor in making their investment decision.

For analyst certification and other important disclosures, refer to the Disclosure Section, located at the end of this report.

+ = Analysts employed by non-U.S. affiliates are not registered with FINRA, may not be associated persons of the member and may not be subject to FINRA restrictions on communications with a subject company, public appearances and trading securities held by a research analyst account.

Contextualizing the AI Transceiver PCB Opportunity

To put the opportunity into perspective, we forecast the global AI transceiver PCB TAM to expand from approximately US\$620mn in CY25 to ~US\$3.8bn by CY28. For comparison, we estimate that Apple (covered by Erik Woodring) – currently the world's largest consumer of HDI PCBs – accounts for roughly US\$3.0-3.5B of annual HDI PCB demand. On our estimates, the AI transceiver PCB market will grow from a niche, sub-US\$1B segment today to a market comparable in size to Apple's HDI PCB demand by CY27, before surpassing it by CY28.

The significance of this comparison extends beyond the absolute market size. For years, Apple has been the primary driver of technology advancement, capacity investment, and profitability within the HDI PCB industry. We believe AI optical connectivity is emerging as the next major demand engine, creating a new growth vector for PCB suppliers and potentially reshaping the industry's competitive landscape.

Given the pace of market expansion, increasing technology requirements, and improving economics associated with next-generation transceiver PCBs, we believe AI transceiver PCBs are rapidly evolving into one of the most compelling structural growth opportunities within the global PCB sector.

Viewed from another angle, Prismark estimates the global HDI PCB market at approximately US\$16.2bn. Based on our forecast of a ~US\$3.8bn AI transceiver PCB market by CY28, AI optical connectivity alone could generate incremental demand equivalent to roughly 23% of the current global HDI market, underscoring the significance of this emerging growth driver for the PCB industry

In February 2026, our global team highlighted that the AI transceiver market is entering an 'exponential' growth phase, driven by scale-out, scale-up and even scale-across in AI datacenters (report link: [AI Transceivers: Growth Dominates Disruption](#)). While CPO (co-packaged optics) remains a fundamental architectural shift and a legitimate long-term risk to pluggable transceivers, the impact on transceivers vendors and their supply chains is limited in the medium term, as large-scale adoption of CPO is unlikely before 2027-28, considering its manufacturing yield, thermal complexity, cost, ecosystem, and serviceability. Thus, our global team expects AI transceiver growth to dominate the disruption from CPO in the medium term, and that transceivers and CPO will coexist through the 3.2T transition expected in CY28.

Our Global team expects strong optical transceiver volumes in CY26-28: In May 2026, our global team raised ([link](#)) its CY26-28 AI transceiver shipments forecast from 53 / 71 / 80mn units to 73 / 141 / 158 M units, respectively, within which 29 / 79 / 87M are 1.6T optical transceivers, with 8mn 3.2T optical transceivers expected in CY28 ([Exhibit 1](#)). This translates into a ~99% CAGR over CY25-28, creating a significant surge in demand for supply chain capacity. This signals a robust growth prospect not only for the optical transceivers vendors but also for the entire supply chain, including the PCB manufacturers. This is because within each optical transceiver module, there is 1 PCB, so it is a one-for-one opportunity for PCB manufacturers.

Exhibit 2: Our Global team raised forecasts for 800G and 1.6T optical transceiver shipments*(mn units)*

New Estimates	2026E	2027E	2028E
800G	44	63	64
1.6T	29	79	87
3.2T	0	0	8
Total	73	141	158

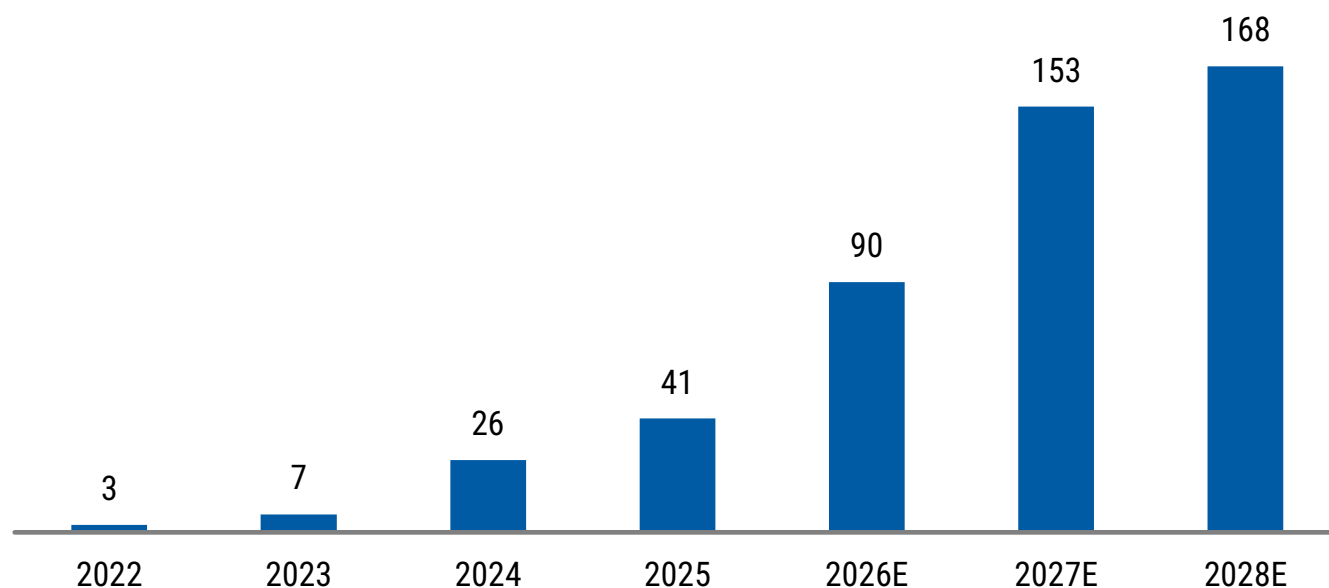
Previous Estimates	2026E	2027E	2028E
800G	34	48	54
1.6T	19	24	27
3.2T	0	0	5
Total	53	71	85

Pct. of change	2026E	2027E	2028E
800G	29%	31%	19%
1.6T	52%	233%	226%
3.2T	--	--	60%
Total	38%	98%	86%

Source: Company data, Morgan Stanley Research estimates.

Exhibit 3: Our Global team estimates AI optical transceivers growing at ~60% CAGR from 2025-28E

We estimate AI transceivers growing at 60% CAGR from 2025-28E



Source: Morgan Stanley Research estimates.

Aside from volume growth, optical transceiver PCB is also seeing content growth...

Our industry checks suggest optical transceiver PCBs will see content growth mainly coming from three aspects.

1. **Upgrades in CCL material (M6 to M8):** The migration from 400G to 800G and 1.6T transceivers is driving increased adoption of higher-specification CCL materials. Whereas 400G modules generally utilized M6-grade CCL, next-generation transceivers increasingly require M7- or M8-grade CCL to meet more demanding signal integrity requirements, supporting higher PCB ASPs.
2. **Increased layer count (10L to 16L):** Increasing PCB layer counts represent another tailwind for ASP growth. Layer requirements rise from 10–12 layers in 400G transceivers to 12–14 layers in 800G and 14–16 layers in 1.6T modules, reflecting greater design complexity and supporting higher PCB ASPs.
3. **More complex manufacturing process (HDI to mSAP):** The adoption of more advanced fabrication technologies is another driver of PCB ASP expansion. While 400G transceivers typically rely on HDI (high density interconnect) processes, next-generation 800G and 1.6T modules increasingly require mSAP (modified semi-additive process), reflecting higher design complexity and supporting higher ASPs.

The combination of the three drivers mentioned above is driving a strong growth of the overall PCB market for optical transceiver modules. The reason why mSAP is required is that it saves space by allowing denser conducting path layouts, which makes it the perfect material for optical transceivers. This is commonly found in compact devices, such as smartphone mainboards and smartphone camera modules. Apple is one of the first companies to start using mSAP technology in its smartphone PCB, going back to 2017, for the iPhone 8/X generation. Which is why today, a lot of the optical transceiver modules makers are also companies that have been HDI/SLP PCB suppliers for Apple iPhone, including Unimicron, ZDT, and Compeq.

...why is why we estimate PCB ASP per transceiver module will rise from US\$5-15 for a 400G transceiver to US\$20-30 for a 1.6T transceiver. We expect the transition from 400G to 1.6T transceivers to drive a substantial increase in PCB content and complexity. PCB designs are likely to upgrade from 10- to 12-layer HDI structures with 2-3 lamination cycles and M6-grade CCL to 14- to 16-layer mSAP-based architectures with 6-10 mSAP layers and M8-grade CCL. Consequently, we estimate PCB ASPs could rise ~2.5x, from approximately US\$10 to US\$25 per unit. Moreover, the higher technical barriers and value-add should drive gross margins from 20-30% to 40-50%+, approaching those of ABF substrates. **The combination of strong volume growth and rising PCB content per transceiver is expected to drive AI transceiver PCB TAM growth well ahead of underlying module shipments.** We estimate the global AI transceiver PCB market will grow at an 83% CAGR over CY25-28E, versus a 60% CAGR growth in AI transceiver unit volumes, reflecting increasing PCB complexity and higher content per module as the industry migrates to 800G and 1.6T transceivers.

Simply put, as the AI transceiver speeds migrate from 400G to 1.6T or 3.2T, the electrical signals need to travel at much higher speeds, and the PCB traces start to make a difference, where signal loss, reflections, crosstalk, and timing variations become major challenges. To preserve signal integrity, the PCB needs to be designed with lower-loss

laminated materials are required (M8 grade CCL), tighter impedance control, more sophisticated routing, optimized vias, etc. As a result, each generation of faster optical transceivers increases the technical complexity and value-added content of the PCB, creating a structural tailwind for advanced PCB suppliers with expertise in mSAP capability.

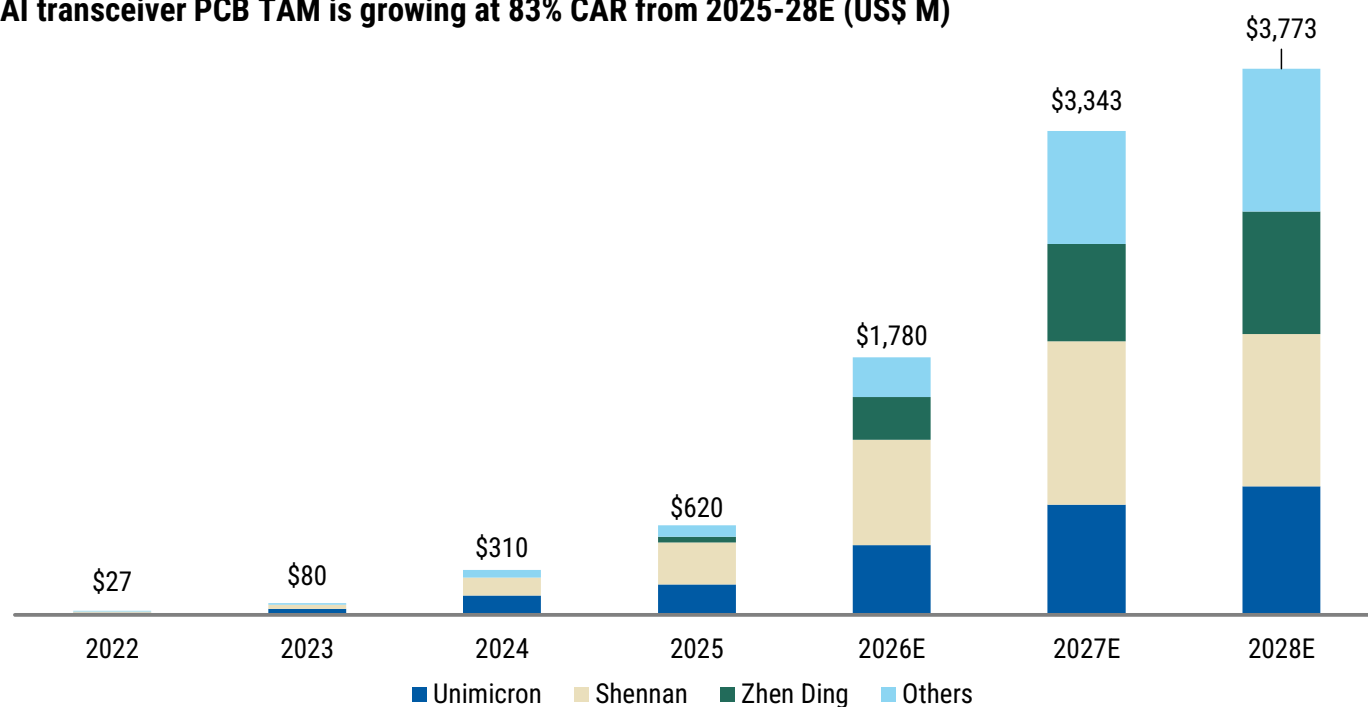
Exhibit 4: Optical transceiver PCB specs

	400G	800G	1.6T
Layers	10-12L 2-3 press HDI	12-14L, 4-8L mSAP or HDI	14-16L 6-10L mSAP
Process	HDI	HDI or mSAP	mSAP
CCL Spec	M6	M7(LDK1) / M7+ (LDK2)	M7+ / M8 (LDK2)
ASP (US\$)	\$5-15	\$15-25	\$20-30
GM Est.	20-30%	30-40%	40-50%+
Suppliers	Unimicron, Shennan, WUS, other smaller suppliers.	Unimicron, Shennan, Zhen Ding, WUS, Compeq, etc.	Unimicron, Shennan, Zhen Ding, Compeq, WUS, etc.

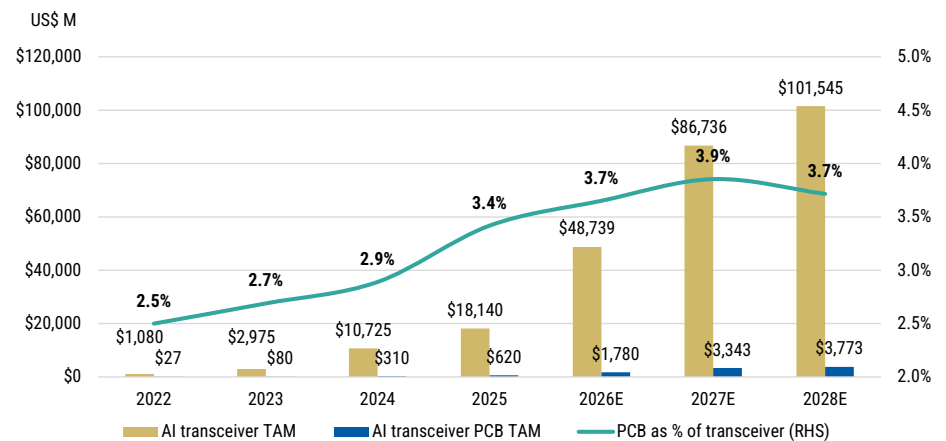
Source: Morgan Stanley Research.

Exhibit 5: We estimate Global AI Transceiver PCB TAM to grow at 83% CAGR from 2025-28E

AI transceiver PCB TAM is growing at 83% CAR from 2025-28E (US\$ M)



Source: Morgan Stanley Research estimates.

Exhibit 6: TAM comparison: AI transceiver vs. AI transceiver PCB

Source: Morgan Stanley Research estimates.

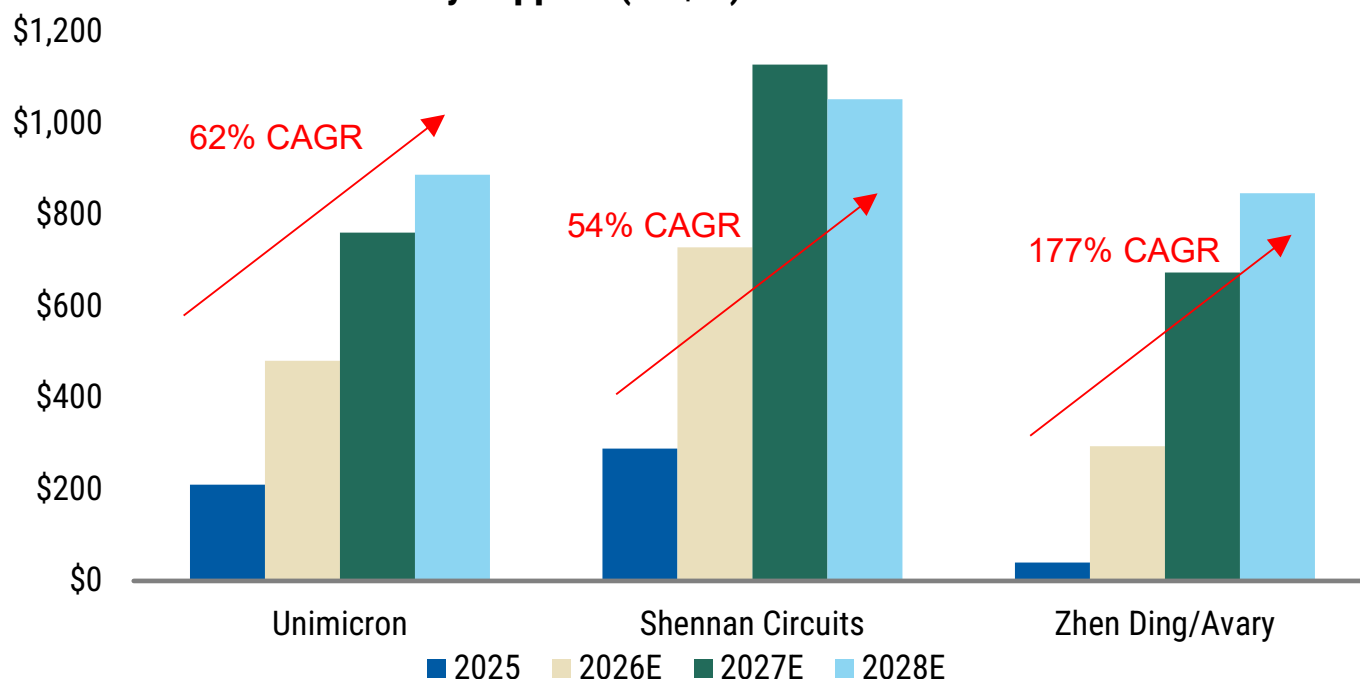
Incumbents like Unimicron and Shennan should continue to do well in the optical transceiver market, but we see ZDT as being aggressive and gaining share: We believe incumbent optical transceiver PCB suppliers, including Shennan and Unimicron, are well positioned to benefit from both volume growth and rising PCB content per transceiver over CY26-28. At the same time, the expanding addressable market is creating opportunities for new entrants, as existing industry capacity appears insufficient to meet anticipated demand.

Our channel checks indicate that ZDT and Compeq (2313.TW, not covered) are actively expanding capacity to support the ramp of 800G and 1.6T optical transceiver PCBs. We believe ZDT's share gains began with the initial ramp-up of 800G transceivers in CY25, and we estimate its supply share increased to approximately 7.5% in CY25, with the potential to reach ~20% by CY28. In the 1.6T segment, we estimate that ZDT entered the market with roughly 5% share in CY25 and could expand its position to ~25% by CY28E.

We expect ZDT's share gains to accelerate with the industry's migration toward 1.6T transceivers and beyond, where PCB technology requirements become increasingly demanding. Specifically, 1.6T+ transceiver PCBs are expected to require mSAP manufacturing processes, raising both technical barriers and qualification requirements. We believe ZDT is particularly well positioned to capitalize on this transition, given its extensive experience with mSAP technology, developed and refined through its participation in Apple's iPhone mainboard supply chain since the adoption of mSAP-based SLP architectures in 2017.

Exhibit 7: AI transceiver PCB value contribution to suppliers - we expect suppliers to see 50-176% CAGR over CY25-28E

AI Transceiver PCB value by supplier (US\$M)



Source: Morgan Stanley Research estimates.

MSe vs. Consensus

Our 2026-28 revenue estimates are 1%, 2% and -3% different from Consensus, while our net income estimates are 0%, 3% and 4% higher than Consensus. We think the variance comes from our incorporation of higher AI transceiver PCB supply share assumptions for ZDT at 17%/20%/22% for 2026/2027/2028. The stock has rallied ~145% since April 1, vs. TAIEX +26% over the same period, driven primarily by the progress of its ABF substrate business and its potential share gains in the transceiver PCB market. Our global team raised its AI transceiver volume forecasts on May 15; thus, we are adjusting our model for ZDT accordingly to reflect the transceiver volume higher estimates.

Although our estimates are not materially above Street expectations, we see scope for further upward revisions as AI demand continues to exceed expectations. We believe technological leadership in mSAP processes, financial capacity to support aggressive expansion, and a willingness to invest ahead of demand will be key differentiators. On these metrics, ZDT appears well positioned to capture outsized share gains as the AI transceiver PCB market develops.

Exhibit 8: Zhen Ding: MSe vs. Consensus

Zhen Ding	MSe			Consensus			Variance		
	2026	2027	2028	2026	2027	2028	2026	2027	2028
Revenue	226,287	291,591	337,349	223,059	285,120	346,952	1%	2%	-3%
GM	24.4%	26.7%	28.8%	23.8%	25.6%	26.9%	1%	1%	2%
OpM	11.5%	14.9%	17.9%	11.2%	14.3%	16.2%	0%	1%	2%
Net income	15,470	25,909	35,713	15,413	25,150	34,349	0%	3%	4%
EPS	14.46	24.21	33.37	14.40	23.50	32.10	0%	3%	4%

Source: Visible Alpha, Morgan Stanley Research estimates.

Our 2026-28 revenue estimates are 2%, 4% and 8% above Consensus, while our net income estimates are 20%, 17% and 8% below Consensus. We think the variance comes from our lower price hikes assumptions for its ABF substrate business compared to the Street's, which resulted in lower margins and profits. We are still of the view that price hikes this year for ABF substrates are primarily to reflect the raw material shortages and cost increases, but heading into 2027, the pricing upsides will be driven by undersupply of substrate capacity itself.

Exhibit 9: Unimicron: MSe vs. Consensus

Unimicron	MSe			Consensus			Variance		
	2026	2027	2028	2026	2027	2028	2026	2027	2028
Revenue	184,777	262,464	354,693	181,665	252,649	328,690	2%	4%	8%
GM	20.0%	26.2%	34.1%	22.7%	31.3%	35.3%	-3%	-5%	-1%
OpM	10.4%	18.6%	27.9%	13.7%	23.4%	27.9%	-3%	-5%	0%
Net income	18,220	38,875	78,482	22,804	47,008	72,397	-20%	-17%	8%
EPS	11.84	25.26	50.99	14.82	30.54	47.04	-20%	-17%	8%

Source: Visible Alpha, Morgan Stanley Research estimates.

Our 2026-28 revenue estimates are 4%, 7% and 13% above Consensus, while our net income estimates are 5% below Consensus for 2026 and 5%/15% above Consensus for 2027/2028. We think the variance comes from our higher assumptions for revenue contribution and profitability from the AI transceiver business, for which Shennan is a share leader. It is primarily valuation that keeps us EW on Shennan.

Exhibit 10: Shennan Circuits: MSe vs. Consensus

Shennan Circuits	MSe			Consensus			Variance		
	2026	2027	2028	2026	2027	2028	2026	2027	2028
Revenue	32,234	42,154	53,710	30,853	39,416	47,667	4%	7%	13%
GM	29.9%	32.1%	33.1%	30.9%	32.2%	32.9%	-3%	-1%	1%
OpM	18.0%	20.7%	22.3%	18.4%	20.2%	21.3%	-2%	3%	5%
Net income	4,785	7,493	10,496	5,020	7,138	9,150	-5%	5%	15%
EPS	7.18	11.24	15.74	7.53	10.71	13.72	-5%	5%	15%

Source: Visible Alpha, Morgan Stanley Research estimates.

Risks to our call

- Weaker demand for general and AI servers;
- Meaningful capacity expansion plans;
- Faster-than-expected adoption of CPO.

ZDT: Estimate Changes

We raise our CY2026, 2027 and 2028 EPS estimates by 5%, 15%, and 17%, respectively. Our earnings increases are driven by our raised demand, pricing, and margin assumptions on AI optical transceiver PCBs.

Exhibit 11: Estimate Changes

NT\$mnn	2026E			2027E			2028E		
	New	Old	Variance	New	Old	Variance	New	Old	Variance
Net sales	226,287	220,957	2%	291,591	272,268	7%	337,349	312,343	8%
COGS	-170,996	-167,319	2%	-213,614	-201,052	6%	-240,162	-224,876	7%
Gross profit	55,291	53,638	3%	77,976	71,216	9%	97,187	87,467	11%
Operating expenses	-29,376	-29,024	1%	-34,573	-33,545	3%	-36,776	-36,126	2%
Operating income	25,914	24,614	5%	43,403	37,671	15%	60,411	51,341	18%
Non-operating income	1,039	1,039	0%	1,406	1,406	0%	1,406	1,406	0%
Pre-tax income	26,953	25,652	5%	44,809	39,077	15%	61,817	52,747	17%
Net income	15,470	14,729	5%	25,909	22,596	15%	35,713	30,487	17%
EPS (NT\$)	14.46	13.76	5%	24.21	21.11	15%	33.37	28.49	17%
Margins (%)			ppt			ppt			ppt
Gross margin	24.4%	24.3%	0.2	26.7%	26.2%	0.6	28.8%	28.0%	0.8
Operating margin	11.5%	11.1%	0.3	14.9%	13.8%	1.0	17.9%	16.4%	1.5
Pre-tax margin	11.9%	11.6%	0.3	15.4%	14.4%	1.0	18.3%	16.9%	1.4
Net margin	6.8%	6.7%	0.2	8.9%	8.3%	0.6	10.6%	9.8%	0.8

Source: Morgan Stanley Research (E) estimates.

Price Target and Valuation Methodology

Our base-case value/price target rises ~17% to NT\$666: It is derived from our multi-stage residual income (RI) model. We use the company's beginning equity plus the present value of all expected residual income of earnings in excess of the cost of capital.

We use the company's beginning equity plus the present value of all expected residual income of earnings in excess of the cost of capital. The RI is positive when ROAE is above the cost of capital and negative when it is below the cost of capital. We assume a cost of equity of 10%, a medium-term growth rate of 15% and a terminal growth rate of 3% (all unchanged).

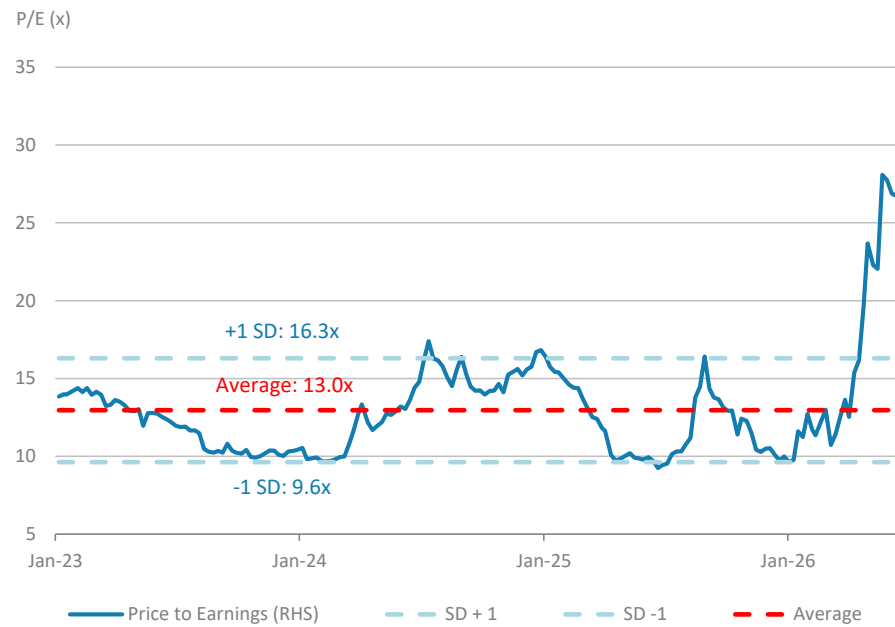
Our bull- and bear-case values also rises by similar magnitudes to NT\$946 and NT\$386, respectively.

Exhibit 12: Residual income (RI) model

NT\$ mn	2026E	2027E	2028E	2029E	2030E	2031E	2032E	2033E	2034E	2035E	2036E
Total Equity	182,543	201,499	225,567	236,401	248,860	263,189	279,666	298,615	320,407	345,467	374,286
Core Net Profit	15,470	25,909	35,713	41,070	47,230	54,315	62,462	71,831	82,606	94,997	109,246
Return on Equity	9.1%	14.2%	17.7%	18.2%	20.0%	21.8%	23.7%	25.7%	27.7%	29.6%	31.6%
Beta (Last 60 Mths)	1.20										
Equity Risk Premium (Rm-Rf)	6%										
Risk Free Rate (Rf)	3%										
Cost of Equity	10%										
Terminal Growth Rate	3%										
Continuing Value Spread	-2%										
Medium-term growth rate	15.0%										
Residual Income	-1,089	8,202	16,167	19,190	24,299	30,175	36,933	44,704	53,640	63,917	75,736
Spread	-1%	4%	8%	9%	10%	12%	14%	16%	18%	20%	22%
Beginning Equity Capital	182,543										
PV of Forecast Period	165,498										
PV of Continuing Value	364,731										
Equity Value	712,772										
No. of Shares	1,070										
Projected Price (EoY)	666.0										
Implied 2027 P/E (x)	27.5										
Implied 2028 P/E (x)	20.0										

Source: Morgan Stanley Research estimates.

Exhibit 13: ZDT P/E



Source: TEJ, Morgan Stanley Research estimates.

Financials

Exhibit 14: Quarterly Estimates

NT\$ mn	1Q25	2Q25	3Q25	4Q25	1Q26	2Q26E	3Q26E	4Q26E	2025	2026E	2027E	2028E
Sales	40,082	38,203	47,366	56,870	40,728	48,174	64,781	72,604	182,522	226,287	291,591	337,349
COGS	-34,197	-31,193	-36,956	-44,040	-31,917	-37,150	-48,224	-53,706	-146,385	-170,996	-213,614	-240,162
Gross profit	5,885	7,011	10,410	12,830	8,812	11,024	16,556	18,899	36,136	55,291	77,976	97,187
Operating expenses	-4,829	-4,586	-5,880	-6,909	-6,308	-7,165	-7,801	-8,103	-22,205	-29,376	-34,573	-36,776
- Promotion	-486	-493	-630	-744	-616	-692	-838	-845	-2,352	-2,990	-3,781	-4,152
- ADM	-1,781	-1,483	-2,270	-2,512	-2,147	-2,489	-2,677	-2,791	-8,046	-10,104	-12,171	-13,367
- R&D	-2,563	-2,611	-2,981	-3,653	-3,545	-3,984	-4,286	-4,467	-11,807	-16,282	-18,621	-19,256
Operating profit	1,056	2,425	4,530	5,921	2,503	3,859	8,756	10,796	13,932	25,914	43,403	60,411
Non-operating income	401	-153	131	-248	-103	314	414	414	131	1,039	1,406	1,406
Interest income	119	91	49	95	64	64	64	64	355	256	256	256
Investment income	-97	203	242	53	2	100	200	200	401	502	550	550
Disposal of investment	0	0	0	0	0	0	0	0	0	0	0	0
Disposal of fixed assets	16	58	-258	-102	0	0	0	0	-286	0	0	0
Exchange gain	103	-103	0	0	-930	0	0	0	0	-930	0	0
Other	259	-401	98	-294	761	150	150	150	-338	1,211	600	600
Pre-tax profit	1,457	2,272	4,661	5,673	2,400	4,173	9,170	11,210	14,063	26,953	44,809	61,817
Income tax	-431	-885	-1,072	-1,070	-353	-918	-1,724	-1,345	-3,458	-4,340	-6,944	-9,605
Net profit	632	605	2,392	3,161	1,426	2,121	4,954	6,970	6,791	15,470	25,909	35,713
EPS (NT\$)	0.66	0.63	2.46	3.22	1.33	1.98	4.63	6.51	6.91	14.46	24.21	33.37
Margins												
Gross margin	15%	18%	22%	23%	22%	23%	26%	26%	20%	24%	27%	29%
Operating margin	3%	6%	10%	10%	6%	8%	14%	15%	8%	11%	15%	18%
Pre-tax margin	4%	6%	10%	10%	6%	9%	14%	15%	8%	12%	15%	18%
Net margin	2%	2%	5%	6%	4%	4%	8%	10%	4%	7%	9%	11%
QoQ Growth												
Sales	-29%	-5%	24%	20%	-28%	18%	34%	12%				
Gross profit	-49%	19%	48%	23%	-31%	25%	50%	14%				
Operating profit	-81%	130%	87%	31%	-58%	54%	127%	23%				
Pre-tax profit	-81%	56%	105%	22%	-58%	74%	120%	22%				
Net profit	-86%	-4%	295%	32%	-55%	49%	134%	41%				
YoY Growth												
Sales	23%	18%	-6%	1%	2%	26%	37%	28%	6%	24%	29%	16%
Gross profit	10%	65%	-9%	12%	50%	57%	59%	47%	11%	53%	41%	25%
Operating profit	42%	-477%	-24%	7%	137%	59%	93%	82%	20%	86%	67%	39%
Pre-tax profit	-3%	346%	-12%	-27%	65%	84%	97%	98%	-7%	92%	66%	38%
Net profit	-35%	25%	-29%	-28%	125%	251%	107%	120%	-26%	128%	67%	38%

Source: Company data, Morgan Stanley Research (E) estimates.

Exhibit 15: Consolidated Financial Summary

Income Statement

NT\$ mn	2025	2026E	2027E	2028E
Net sales	182,522	226,287	291,591	337,349
COGS	-146,385	-170,996	-213,614	-240,162
Gross profit	36,136	55,291	77,976	97,187
Operating expenses	-22,205	-29,376	-34,573	-36,776
- Promotion	-2,352	-2,990	-3,781	-4,152
- ADM	-8,046	-10,104	-12,171	-13,367
- R&D	-11,807	-16,282	-18,621	-19,256
Operating income	13,932	25,914	43,403	60,411
Non-operating income	131	1,039	1,406	1,406
Interest income	355	256	256	256
Investment income	401	502	550	550
Disposal of investment	0	0	0	0
Disposal of fixed assets	-286	0	0	0
Exchange gain	0	-930	0	0
Other	-338	1,211	600	600
Pre-tax income	14,063	26,953	44,809	61,817
Income tax	-7,272	-11,483	-18,900	-26,104
Net income	6,791	15,470	25,909	35,713
EPS (NT\$)	6.91	14.46	24.21	33.37

Balance Sheet

NT\$ mn	2025	2026E	2027E	2028E
Cash	71,116	61,840	73,189	88,287
Mkt securities	36	36	36	36
Accounts/Notes receivables	31,412	38,944	50,183	58,058
Inventory	19,616	22,914	28,625	32,183
Other current assets	5,460	5,460	5,460	5,460
Current Assets	127,641	129,195	157,494	184,024
Long-term investments	9,866	10,367	10,917	11,467
Fixed assets	123,737	145,693	148,605	152,383
Other assets	18,799	18,799	18,799	18,799
Total Assets	280,043	304,054	335,815	366,673
S/T borrowings	23,839	23,839	23,839	23,839
AP/NP	23,501	27,452	34,295	38,557
Other ST liabilities	25,481	33,710	39,674	42,202
Other liabilities	21,151	21,151	21,151	21,151
L/T debt	15,359	15,359	15,359	15,359
Total Liabilities	109,330	121,511	134,316	141,106
Common shares	10,706	10,706	10,706	10,706
Capital Collection	0	0	0	0
APIC	52,902	52,902	52,902	52,902
Retained earnings	61,549	73,328	92,283	116,351
Treasury Stock	-52	0	0	0
Other shareholders' equity	45607	45607	45607	45607
Shareholders' equity	170,713	182,543	201,499	225,567
Total Liab./Shrholder's Equity	280,043	304,054	335,815	366,673

Cash Flow Statement

NT\$ mn	2025	2026E	2027E	2028E
Cashflow from operations	27,617	24,364	28,302	36,742
Net Profits	10,605	15,470	25,909	35,713
Depreciation	18,552	8,044	7,088	6,222
Equity investment losses (income)	0	-502	-550	-550
Other adjustments	-1,541	1,351	-4,145	-4,643
Cashflow from investing	-31,960	-30,000	-10,000	-10,000
(Purchases) sale of FA (capex)	-32,673	-30,000	-10,000	-10,000
(Purchases) sale of L/T investment	342	0	0	0
(Purchases) sale of S/T investment	-601	0	0	0
Other adjustments	972	0	0	0
Cashflow from financing	-1,865	-3,640	-6,953	-11,645
Increase in L/T debt	6,344	0	0	0
Increase in S/T debt	2,191	0	0	0
Issuance of stock	730	0	0	0
Cash dividends	-7,221	-3,692	-6,953	-11,645
Dir. & Emp. Bonus	0	0	0	0
Changes in Treasury Stocks	288	52	0	0
Other adjustments	-4,196	0	0	0
Exchange rate adjustment	-2,177	0	0	0
Net change in cash	-8,386	-9,276	11,349	15,097

Financial Ratios

	2025	2026E	2027E	2028E
Margins				
Gross margin	19.8%	24.4%	26.7%	28.8%
Operating margin	7.6%	11.5%	14.9%	17.9%
Pretax margin	7.7%	11.9%	15.4%	18.3%
Net margin	3.7%	6.8%	8.9%	10.6%
YoY growth				
Sales	6.3%	24.0%	28.9%	15.7%
Operating profits	20.2%	86.0%	67.5%	39.2%
Pretax profits	-6.5%	91.7%	66.2%	38.0%
Net profits	-26.0%	127.8%	67.5%	37.8%
EPS	-28.6%	109.3%	67.5%	37.8%
Net Debt/Equity (Net of mkt secs.)	-19%	-12%	-17%	-22%
Net Debt/Equity	-19%	-12%	-17%	-22%
Liabilities/Equity	64%	67%	67%	63%
Liabilities/Assets	39%	40%	40%	38%
ROAE	4.2%	8.8%	13.5%	16.7%
ROAA	2.5%	5.3%	8.1%	10.2%
AR/NR Turnover (days)	62	57	56	59
AP/NP Turnover (days)	56	54	53	55
Inventory Turnover (days)	47	45	44	46
Cash conversion cycle (days)	52	48	47	49

Source: Company data, Morgan Stanley Research (E) estimates.

Unimicron: Estimate Changes

We raise our 2026, 2027 and 2028 EPS estimates by 2%, 5% and 4%, respectively, primarily driven by higher assumptions for AI optical transceiver PCBs.

Exhibit 16: Estimate Changes

Year to Dec. 31 (NT\$m)	2026E			2027E			2028E		
	New	Old	% diff	New	Old	% diff	New	Old	% diff
P&L Summary									
Net sales	184,777	186,189	-1%	262,464	260,849	1%	354,693	352,349	1%
COGS	(147,780)	(149,522)		(193,803)	(194,590)		(233,768)	(235,051)	
Gross profit	36,997	36,667	1%	68,660	66,260	4%	120,924	117,299	3%
Operating expenses	(17,713)	(17,794)		(19,966)	(19,966)		(21,971)	(22,086)	
Operating income	19,284	18,872	2%	48,695	46,293	5%	98,953	95,212	4%
Non-operating income	5,027	5,027		2,180	2,180		2,180	2,180	
Pre-tax income	24,311	23,899	2%	50,874	48,473	5%	101,133	97,392	4%
Net income	18,220	17,879	2%	38,875	36,979	5%	78,482	75,534	4%
Diluted EPS (NT\$)	11.84	11.62	2%	25.26	24.03	5%	50.99	49.08	4%
Margins									
Gross margin	20.0%	19.7%	ppt	26.2%	25.4%	ppt	34.1%	33.3%	ppt
Operating margin	10.4%	10.1%	0.3	18.6%	17.7%	0.8	27.9%	27.0%	0.9
Pretax margin	13.2%	12.8%	0.3	19.4%	18.6%	0.8	28.5%	27.6%	0.9
Net margin	9.9%	9.6%	0.3	14.8%	14.2%	0.6	22.1%	21.4%	0.7

Source: Morgan Stanley Research (E) estimates.

Price Target and Valuation Methodology

Our 12-month price target increases ~5% to NT\$1,285 (from NT\$1,225), driven by our increased earnings estimates for CY26-28e: We believe the company's earnings have returned to a growth track from 2026 onwards and will see stronger earnings growth in 2027, driven by the high-end ABF substrate market's reversion to undersupply and Unimicron's dominant supply share in AI ASICs and also server CPUs, which puts it in a favorable position to benefit from rising AI investments.

We believe the overall ABF substrate market bottomed in 2H24, and growth will accelerate from 2H26 onwards with the ramp of multiple next-generation, large body size ABF substrates for new server GPUs, CPUs, ASICs, and networking chips. New technologies, such as embedded passives and Intel's EMIB-T, will also increase the manufacturing complexity of the package substrates, which adds to the under-capacity as usable output decreases.

Similar to our analysis of other tech hardware companies within our coverage, we use a residual income (RI) valuation model to value Unimicron – we think it derives the most accurate value of the firm as it takes into account cost of equity. Key assumptions are all unchanged, including:

- Cost of equity of 9.2%;
- Risk-free rate of 1% (10-year Taiwan government note yield);
- Equity risk premium of 8.7%;
- Beta of 1.0;
- Medium-term growth rate of 15%
- Terminal growth rate of 3% (similar to the rest of the companies under our tech hardware coverage).

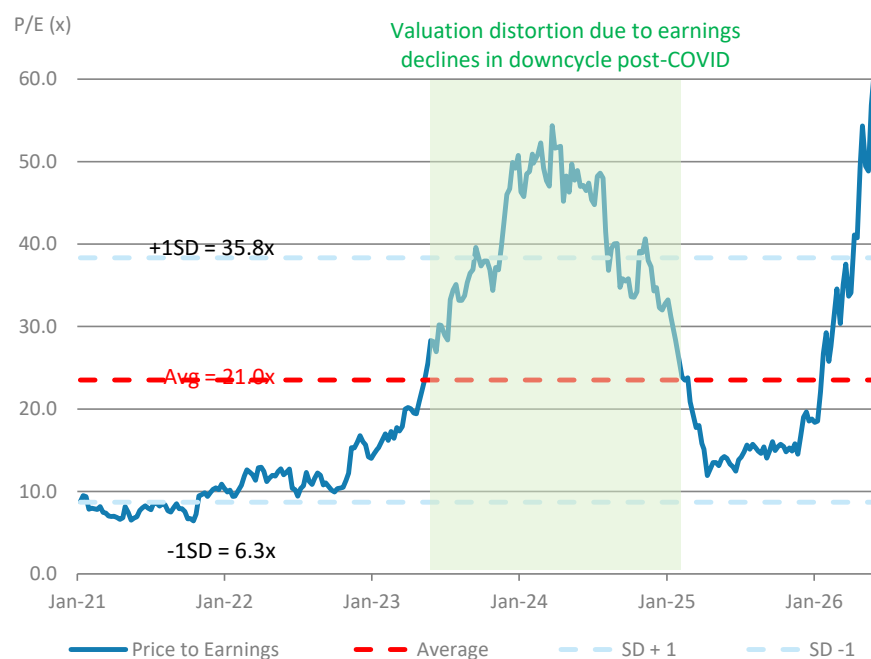
Our bull and bear case values increase by similar magnitudes, to NT\$2,190 and NT\$430, respectively.

Exhibit 17: Unimicron residual income (RI) model

	2027E	2028E	2029E	2030E	2031E	2032E	2033E	2034E	2035E	2036E	2037E
Total Equity	152,160	212,630	246,731	285,946	331,044	382,906	442,548	511,136	590,012	680,719	756,223
Net Profit	38,875	78,482	90,254	103,792	119,361	137,265	157,855	181,533	208,763	240,077	247,280
Return on Equity	28.4%	43.0%	39.3%	39.0%	38.7%	38.5%	38.2%	38.1%	37.9%	37.8%	34.4%
Beta	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
Equity Risk Premium (Rm-Rf)	8.7%	8.7%	8.7%	8.7%	8.7%	8.7%	8.7%	8.7%	8.7%	8.7%	8.7%
Risk Free Rate (Rf)	1.0%	1.0%	1.0%	1.0%	1.0%	1.0%	1.0%	1.0%	1.0%	1.0%	1.0%
Cost of Equity	9.2%	9.2%	9.2%	9.2%	9.2%	9.2%	9.2%	9.2%	9.2%	9.2%	9.2%
Residual Income	23,387	51,508	64,040	73,507	84,394	96,912	111,308	127,863	146,900	168,792	171,813
Spread	19.2%	33.9%	30.1%	29.8%	29.5%	29.3%	29.1%	28.9%	28.7%	28.6%	25.2%
Beginning Equity Capital	152,160										
PV of Forecast Period	634,902										
PV of Continuing Value	1,190,556										
Equity Value	1,977,619										
No. of Shares	1,539										
Projected Price	1,285.0										

Source: Morgan Stanley Research estimates.

Exhibit 18: Unimicron P/E



Source: TEJ, Morgan Stanley Research estimates.

Financials

Exhibit 19: Quarterly Estimates

NT\$ mn	1Q25	2Q25	3Q25	4Q25	1Q26E	2Q26E	3Q26E	4Q26E	2025	2026E	2027E	2028E
Sales	30,090	32,466	33,994	34,691	37,446	43,621	51,100	52,609	131,241	184,777	262,464	354,693
Cost of goods sold	-26,063	-28,221	-29,446	-29,219	-30,720	-35,125	-40,558	-41,377	-112,949	-147,780	-193,803	-233,768
Gross profit	4,026	4,246	4,549	5,472	6,726	8,497	10,542	11,232	18,292	36,997	68,660	120,924
Operating expenses	-2,759	-2,736	-3,016	-3,106	-3,969	-4,331	-4,696	-4,717	-11,618	-17,713	-19,966	-21,971
Operating profit	1,267	1,509	1,533	2,366	2,757	4,166	5,846	6,516	6,674	19,284	48,695	98,953
Interest income (loss)	103	62	-7	25	25	25	25	25	183	100	100	100
Investment income (loss)	32	-13	-1	-10	20	20	20	20	8	80	80	80
Gains (Losses) from investment disposal	0	0	0	3	0	0	0	0	3	0	0	0
Gains from FAs disposal	3	337	-29	507	0	0	0	0	817	0	0	0
Exchange gain	452	-1,281	487	349	300	0	0	0	7	300	0	0
Others	-483	-257	964	910	3,197	450	450	450	1,134	4,547	2,000	2,000
Pre-tax profit	1,373	357	2,947	4,150	6,299	4,661	6,341	7,010	8,827	24,311	50,874	101,133
Net Income for Minority	9	231	256	381	338	338	338	338	877	1,351	1,351	1,351
Income tax	-450	-97	-496	-235	-918	-1,165	-1,395	-1,262	-1,277	-4,740	-10,648	-21,300
Net profit	915	30	2,194	3,535	5,043	3,158	4,608	5,411	6,673	18,220	38,875	78,482
EPS (NT\$)	0.59	0.02	1.43	2.30	3.28	2.05	2.99	3.52	4.34	11.84	25.26	50.99
Margins (%)												
Gross margin	13.4%	13.1%	13.4%	15.8%	18.0%	19.5%	20.6%	21.4%	13.9%	20.0%	26.2%	34.1%
Operating margin	4.2%	4.6%	4.5%	6.8%	7.4%	9.6%	11.4%	12.4%	5.1%	10.4%	18.6%	27.9%
Pre-tax margin	4.6%	1.1%	8.7%	12.0%	16.8%	10.7%	12.4%	13.3%	6.7%	13.2%	19.4%	28.5%
Net margin	3.0%	0.1%	6.5%	10.2%	13.5%	7.2%	9.0%	10.3%	5.1%	9.9%	14.8%	22.1%
QoQ Growth												
Sales	2%	8%	5%	2%	8%	16%	17%	3%				
Gross profit	18%	5%	7%	20%	23%	26%	24%	7%				
Operating profit	86%	19%	2%	54%	17%	51%	40%	11%				
Pre-tax profit	331%	-74%	726%	41%	52%	-26%	36%	11%				
Net profit	1531%	-97%	7310%	61%	43%	-37%	46%	17%				
YoY Growth												
Sales	14%	16%	7%	18%	24%	34%	50%	52%	14%	41%	42%	35%
Gross profit	-6%	15%	-8%	61%	67%	100%	132%	105%	12%	102%	86%	76%
Operating profit	-20%	68%	-22%	248%	118%	176%	281%	175%	30%	189%	153%	103%
Pre-tax profit	-59%	-83%	95%	1201%	359%	1207%	115%	69%	21%	175%	109%	99%
Net profit	-62%	-98%	120%	6204%	451%	10567%	110%	53%	31%	173%	113%	102%

Source: Company data, Morgan Stanley Research (E) estimates.

Exhibit 20: Consolidated Financial Summary**Income Statement**

NT\$ mn; FY End Dec	2025	2026E	2027E	2028E
Net sales	131,241	184,777	262,464	354,693
Cost of goods sold	-112,949	-147,780	-193,803	-233,768
Gross profit	18,292	36,997	68,660	120,924
Operating expenses	-11,618	-17,713	-19,966	-21,971
Sales & Marketing	-1,607	-2,514	-3,249	-3,994
G&A	-5,012	-7,806	-8,136	-8,550
R&D	-4,999	-7,392	-8,581	-9,428
Operating profit	6,674	19,284	48,695	98,953
Interest income	183	100	100	100
Investment income	8	80	80	80
Gains from investment disposal	3	0	0	0
Gains from fixed assets disposal	817	0	0	0
Exchange gain	7	300	0	0
Other non-operating income	1,134	4,547	2,000	2,000
Net non-operating profit	2,153	5,027	2,180	2,180
Pre-tax profit	8,827	24,311	50,874	101,133
Net Income for Minority	-877	-1,351	-1,351	-1,351
Income tax	-1,277	-4,740	-10,648	-21,300
Net profit	6,673	18,220	38,875	78,482
Reported EPS (NT\$)	4.34	11.84	25.26	50.99

Balance Sheet

NT\$ mn; FY End Dec	2025	2026E	2027E	2028E
Cash	54,872	81,793	141,429	229,011
Mkt Securities	38	38	38	38
AR/NR	27,570	38,816	55,136	74,511
Inventory	17,802	23,292	30,545	36,844
Others	5,016	7,062	10,031	13,556
Current Assets	105,298	151,001	237,179	353,960
Long-term Investments	836	836	836	836
Fixed Assets	125,866	129,535	121,838	114,413
Other Assets	23,792	23,792	23,792	23,792
Total Assets	255,792	305,164	383,645	493,001
S-t Borrowings	10,845	10,845	10,845	10,845
AP/NP	17,494	22,889	30,018	36,208
Other S-T Liabilities	47,115	61,645	80,843	97,514
Other LT Liabilities	38,457	54,144	76,908	103,934
L-t Debt	34,873	33,872	32,871	31,870
Total Liabilities	148,785	183,395	231,485	280,371
Common Shares	15,296	15,296	15,296	15,296
Other Shareholders' Equity	91,711	106,473	136,864	197,334
Total Equity	107,007	121,769	152,160	212,630
Total Liab./Shrhldr's Equity	255,792	305,164	383,645	493,001

Cash Flow Statement

NT\$ mn; FY End Dec	2025	2026E	2027E	2028E
Cashflow from Operations	14,967	39,169	58,657	90,949
Net Profits	6,673	18,220	38,875	78,482
Depreciation	18,773	19,887	20,078	18,885
Investment losses (income)	0	-80	-80	-80
Investment Disposal Loss	-3	0	0	0
Change in Working Capital	-2,943	3,188	2,753	-2,812
Other adjustments	-7,533	-2,046	-2,969	-3,525
Cashflow from Investing	-23,018	-23,556	-12,381	-11,460
(Purchases) Sale of fixed asset	-24,307	-23,556	-12,381	-11,460
(Purchases) Sale of LT investment	-77	0	0	0
Other adjustments	1,367	0	0	0
Cashflow from financing	19,073	11,608	13,359	8,093
Increase in L-T debt	9,950	-1,001	-1,001	-1,001
Increase in S-T debt	4,331	0	0	0
Issuance of stock	0	0	0	0
Other adjustments	7,087	15,687	22,764	27,025
Exchange Rate Adjustment	96	-300	0	0
Net change in cash	11,118	26,921	59,636	87,582

Financial Ratios

	2025	2026E	2027E	2028E
Margins				
Gross margin	13.9%	20.0%	26.2%	34.1%
Operating expenses/sales	-8.9%	-9.6%	-7.6%	-6.2%
Operating margin	5.1%	10.4%	18.6%	27.9%
Tax rate	14.5%	19.5%	20.9%	21.1%
Pre-tax margin	6.7%	13.2%	19.4%	28.5%
Net Margin	5.1%	9.9%	14.8%	22.1%
YoY Growth				
Turnover	13.8%	40.8%	42.0%	35.1%
Gross Profit	12.1%	102.3%	85.6%	76.1%
Operating Profits	30.4%	188.9%	152.5%	103.2%
Pretax Profits	20.6%	175.4%	109.3%	98.8%
Net Profits	31.3%	173.0%	113.4%	101.9%
Adjusted Cash Dividend (NT\$)	2.00	5.46	11.65	23.52
Net Debt/Equity	-8.6%	-30.5%	-64.2%	-87.6%
Liabilities/Equity	139.0%	150.6%	152.1%	131.9%
Liabilities/Assets	58.2%	60.1%	60.3%	56.9%
ROAE	6.5%	15.9%	28.4%	43.0%
ROAA	2.7%	6.5%	11.3%	17.9%
A/R Turnover (days)	71.2	65.6	65.3	66.7
A/P Turnover (days)	52.7	49.9	49.8	51.7
Inventory Turnover (days)	52.8	50.7	50.7	52.6
Cash conversion (days)	71.3	66.4	66.2	67.6

Source: Company data, Morgan Stanley Research (E) estimates.

Shennan Circuits: Estimate Changes

We raise our 2026, 2027 and 2028 EPS estimates by 9%, 17% and 24%, respectively, primarily driven by higher assumptions for AI optical transceiver PCBs.

Exhibit 21: Estimate Changes

Year to Dec. 31 (RMB mn)	2026E			2027E			2028E		
	New	Old	% diff	New	Old	% diff	New	Old	% diff
P&L Summary									
Net sales	32,234	30,721	5%	42,154	37,580	12%	53,710	44,952	19%
COGS	(22,602)	(21,691)		(28,642)	(25,773)		(35,954)	(30,356)	
Gross profit	9,632	9,030	7%	13,512	11,807	14%	17,756	14,596	22%
Operating expenses	(3,843)	(3,664)		(4,794)	(4,277)		(5,752)	(4,821)	
Operating income	5,789	5,366	8%	8,717	7,530	16%	12,004	9,775	23%
Non-operating income	(531)	(531)		(515)	(515)		(515)	(515)	
Pre-tax income	5,258	4,835	9%	8,202	7,015	17%	11,489	9,260	24%
Income tax	(472)	(435)		(710)	(606)		(993)	(799)	
Minority interest	(1)	(1)		-	-		-	-	
Net income	4,785	4,399	9%	7,493	6,409	17%	10,496	8,461	24%
EPS (RMB)	7.18	6.60	9%	11.24	9.61	17%	15.74	12.69	24%
Margins									
			<i>ppt</i>			<i>ppt</i>			<i>ppt</i>
Gross margin	29.9%	29.4%	0.5	32.1%	31.4%	0.6	33.1%	32.5%	0.6
Operating margin	18.0%	17.5%	0.5	20.7%	20.0%	0.6	22.3%	21.7%	0.6
Pretax margin	16.3%	15.7%	0.6	19.5%	18.7%	0.8	21.4%	20.6%	0.8
Net margin	14.8%	14.3%	0.5	17.8%	17.1%	0.7	19.5%	18.8%	0.7

Source: Morgan Stanley Research (E) estimates.

Price Target and Valuation Methodology

We raise our 12-month price target to Rmb400 (implies 36x 2027e EPS or 25x 2028e EPS), up from Rmb320, primarily driven by the increases in our earnings estimates for CY26, CY27 and CY28: Our base case scenario value is derived from a residual income (RI) model.

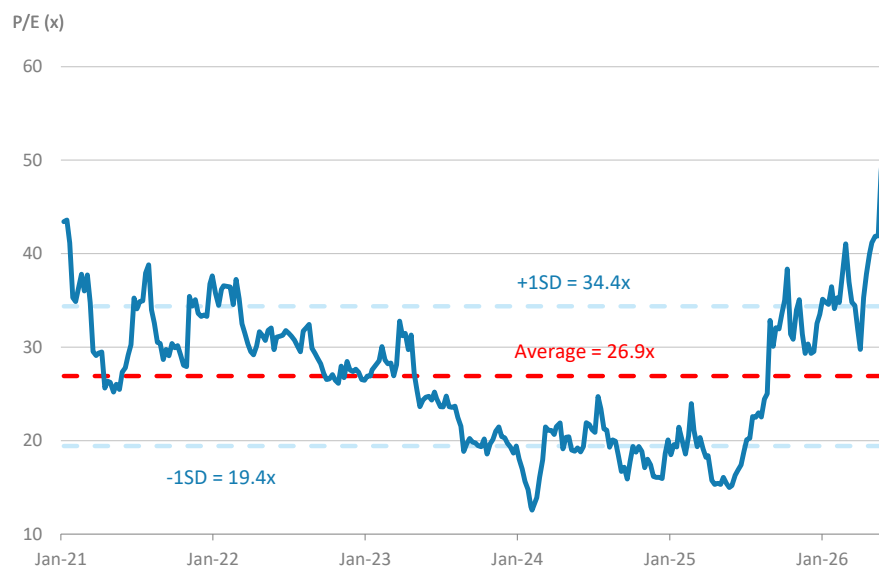
Our key assumptions are all unchanged, including a cost of equity of 6.7% (beta of 1.1, equity risk premium 5.0%, and risk-free rate of 1.3%), a medium-term growth rate of 7.0%, and a terminal growth rate of 4.0%.

Our bull and bear case values increase to Rmb555 and Rmb230, respectively: This is in line with the magnitude of increase in our base case value.

Exhibit 22: Shennan Circuits residual income (RI) model

RMB mn	2026E	2027E	2028E	2029E	2030E	2031E	2032E	2033E	2034E	2035E	2036E
Total Equity	20,731	26,645	34,668	43,604	53,164	63,394	74,340	86,053	98,585	111,994	126,342
Net Profit	4,785	7,493	10,496	11,231	12,017	12,858	13,758	14,721	15,752	16,854	18,034
Return on Equity	25.2%	31.6%	34.2%	28.7%	24.8%	22.1%	20.0%	18.4%	17.1%	16.0%	15.1%
Beta	1.1	1.1	1.1	1.1	1.1	1.1	1.1	1.1	1.1	1.1	1.1
Equity Risk Premium (Rm-Rf)	5.0%	5.0%	5.0%	5.0%	5.0%	5.0%	5.0%	5.0%	5.0%	5.0%	5.0%
Risk Free Rate (Rf)	1.3%	1.3%	1.3%	1.3%	1.3%	1.3%	1.3%	1.3%	1.3%	1.3%	1.3%
Cost of Equity	6.7%	6.7%	6.7%	6.7%	6.7%	6.7%	6.7%	6.7%	6.7%	6.7%	6.7%
Beginning Equity Capital	20,731										
PV of Forecast Period	56,125										
PV of Continuing Value	189,572										
Equity Value	266,727										
No. of Shares	667										
Projected Price (RMB)	400.0										

Source: Morgan Stanley Research estimates.

Exhibit 23: Shennan Circuits 5Y P/E Band

Source: TEJ, Morgan Stanley Research estimates.

Financials

Exhibit 24: Quarterly Estimates

RMB mn	1Q25	2Q25	3Q25	4Q25	1Q26	2Q26E	3Q26E	4Q26E	2025A	2026E	2027E	2028E
Net sales	4,783	5,671	6,301	6,893	6,596	7,780	8,612	9,247	23,647	32,234	42,154	53,710
COGS	(3,621)	(4,144)	(4,364)	(4,947)	(4,706)	(5,478)	(5,986)	(6,432)	(17,076)	(22,602)	(28,642)	(35,954)
Gross profit	1,162	1,527	1,937	1,946	1,889	2,301	2,627	2,815	6,571	9,632	13,512	17,756
Operating expenses	(622)	(668)	(860)	(887)	(804)	(934)	(1,033)	(1,073)	(3,038)	(3,843)	(4,794)	(5,752)
SG&A, R&D exp.	(622)	(668)	(860)	(887)	(804)	(934)	(1,033)	(1,073)	(3,038)	(3,843)	(4,794)	(5,752)
Employee bonus	13	14	15	16	17	18	19	20	1	2	3	4
Operating income	540	858	1,077	1,058	1,085	1,368	1,593	1,743	3,533	5,789	8,717	12,004
Non-operating income	(17)	117	(17)	6	(145)	(129)	(129)	(129)	91	(531)	(515)	(515)
Interest income	(131)	(93)	(116)	(82)	(179)	(179)	(179)	(179)	(422)	(715)	(715)	(715)
Investment income	3	0	(0)	0	(0)	(0)	(0)	(0)	3	(0)	(0)	(0)
Disposal of investment	0	0	0	0	0	0	0	0	0	0	0	0
Exchange gain	0	0	0	0	0	0	0	0	0	0	0	0
Other	111	210	100	88	34	50	50	50	509	184	200	200
Pre-tax income	523	976	1,060	1,065	941	1,239	1,464	1,614	3,624	5,258	8,202	11,489
Income tax	(31)	(106)	(93)	(114)	(89)	(136)	(117)	(129)	(345)	(472)	(710)	(993)
Minority interest	(1)	(0)	(1)	(0)	(1)	0	0	0	(3)	(1)	0	0
Net income	491	869	966	950	850	1,103	1,347	1,485	3,276	4,785	7,493	10,496
Adj.wtd.avg.shrs (mn)	667	667	667	667	667	667	667	667	667	667	667	667
EPS (RMB)	0.74	1.30	1.45	1.42	1.28	1.65	2.02	2.23	4.91	7.18	11.24	15.74
Fully diluted shares (mn)	667	667	667	667	667	667	667	667	667	667	667	667
Diluted EPS (RMB)	0.74	1.30	1.45	1.42	1.28	1.65	2.02	2.23	4.91	7.18	11.24	15.74
Margins (%)												
Gross Margin	24.3	26.9	30.7	28.2	28.6	29.6	30.5	30.4	27.8	29.9	32.1	33.1
Operating Margin	11.3	15.1	17.1	15.4	16.5	17.6	18.5	18.8	14.9	18.0	20.7	22.3
Pretax Margin	10.9	17.2	16.8	15.4	14.3	15.9	17.0	17.5	15.3	16.3	19.5	21.4
Net Margin	10.3	15.3	15.3	13.8	12.9	14.2	15.6	16.1	13.9	14.8	17.8	19.5
QoQ Growth (%)												
Sales	-2	19	11	9	-4	18	11	7				
Gross Profit	12	31	27	0	-3	22	14	7				
Operating Profit	47	59	25	-2	3	26	16	9				
Pretax Profit	24	86	9	0	-12	32	18	10				
Net Profit	26	77	11	-2	-10	30	22	10				
YoY Growth (%)												
Sales	21	30	33	42	38	37	37	34	32	36	31	27
Gross Profit	20	33	66	87	63	51	36	45	52	47	40	31
Operating Profit	30	41	73	188	101	59	48	65	75	64	51	38
Pretax Profit	29	46	102	151	80	27	38	52	79	45	56	40
Net Profit	29	43	93	144	73	27	39	56	74	46	57	40

Source: Company data, Morgan Stanley Research (E) estimates.

Exhibit 25: Consolidated Financial Summary**Consolidated Income Statement**

RMBmn (Year End Dec 31)	2025A	2026E	2027E	2028E
Net sales	23,647	32,234	42,154	53,710
COGS	(17,076)	(22,602)	(28,642)	(35,954)
Gross profit	6,571	9,632	13,512	17,756
Operating expenses	(3,038)	(3,843)	(4,794)	(5,752)
Operating income	3,533	5,789	8,717	12,004
Non-operating income	91	(531)	(515)	(515)
Interest income	(422)	(715)	(715)	(715)
Investment income	3	(0)	(0)	(0)
Disposal of investment	-	-	-	-
Exchange gain	-	-	-	-
Other	509	184	200	200
Pre-tax income	3,624	5,258	8,202	11,489
Income tax	(345)	(472)	(710)	(993)
Minority interests	(3)	(1)	-	-
Net income	3,276	4,785	7,493	10,496
Adj. wtd. Avg. shrs (m)	667	667	667	667
Reported EPS (Rmb)	4.91	7.18	11.24	15.74
Diluted shrs (m)	667	667	667	667
Diluted EPS (Rmb)	4.91	7.18	11.24	15.74

Consolidated Balance Sheet

RMBmn (Year End Dec 31)	2025A	2026E	2027E	2028E
Cash	997	4,694	5,048	5,850
Mkt securities	-	-	-	-
AR/NR	6,669	3,546	4,637	5,908
Inventory	5,140	3,390	4,296	5,393
Others	849	161	211	269
Current Assets	13,654	11,791	14,192	17,419
Long-term investments	3	3	3	3
Fixed assets	15,190	20,315	27,017	35,556
Other assets	1,736	1,736	1,736	1,736
Total Assets	30,583	33,845	42,948	54,714
S/T borrowings	15	515	1,015	1,515
AP/NP	5,625	3,164	4,010	5,033
Other ST liabilities	4,384	5,803	7,353	9,230
Total Current Liabilities	10,024	9,482	12,378	15,779
L/T debt	2,679	2,679	2,679	2,679
Other LT liabilities	699	953	1,246	1,588
Total Liabilities	13,402	13,114	16,303	20,046
Common shares	667	513	513	513
Retained Earnings	10,024	13,728	19,642	27,666
Other SH' Equity	6,490	6,490	6,490	6,490
Total Shareholders' Equity	17,181	20,731	26,645	34,668
Total Liab./SH's Equity	30,583	33,845	42,948	54,714

Consolidated Cash Flow Statement

RMBmn (Year End Dec 31)	2025A	2026E	2027E	2028E
Operating Cashflow	3,838	9,303	7,842	10,971
Net Profits	3,276	4,785	7,493	10,496
Depreciation & Amort.	0	0	0	0
Investment losses/(income)	(3)	0	0	0
Working capital change	(802)	3,830	399	533
Other adjustments	1,368	688	(50)	(58)
Investing Cashflow	(3,756)	(5,125)	(6,702)	(8,539)
Capex	(3,765)	(5,133)	(6,712)	(8,553)
Change of L/T investment	0	0	0	0
Change of S/T investment	0	0	0	0
Other adjustments	4	0	0	0
Financing Cashflow	(650)	(481)	(786)	(1,631)
Increase in L/T debt	207	0	0	0
Increase in S/T debt	0	500	500	500
Issuance of stock	0	(154)	0	0
Cash dividends	(846)	(1,081)	(1,579)	(2,473)
Other adjustments	(11)	254	293	342
FX adjustment	12	0	0	0
Net change in cash	(555)	3,697	355	801

Consolidated Financial Ratios

	2025A	2026E	2027E	2028E
Margins (%)				
Gross margin	27.8	29.9	32.1	33.1
Operating margin	14.9	18.0	20.7	22.3
Pretax margin	15.3	16.3	19.5	21.4
Net margin	13.9	14.8	17.8	19.5
YoY growth (%)				
Sales	32.1	36.3	30.8	27.4
Operating profits	75.4	63.8	50.6	37.7
Pretax profits	79.1	45.1	56.0	40.1
Net profits	74.5	46.1	56.6	40.1
Others				
Cash dividend payout (%)	33	33	33	-
Cash div (Rmb)	1.6	2.4	3.7	5.2
Yield (%)	1	1	2	3
Net Debt/Equity (%)	10	(7)	(5)	(5)
Liabilities/Equity (%)	78	63	61	58
Liabilities/Assets (%)	44	39	38	37
ROAE (%)	21	25	32	34
ROAA (%)	12	15	20	21
AR/NR Turnover (days)	91	58	35	36
Inventory Turnover (days)	91	69	49	49
AP/NP Turnover (days)	107	71	46	46
Cash conversion cycle	75	56	39	39

Source: Company data, Morgan Stanley Research (E) estimates.

Risk Reward – Zhen Ding (4958.TW)

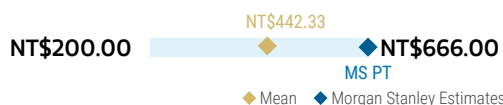
Growing China & International AI chip substrate and PCB vendor

PRICE TARGET NT\$666.00

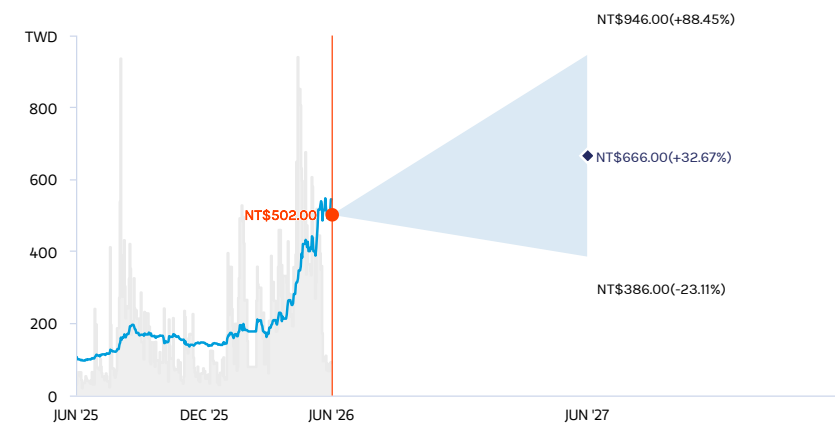
Base case, residual income model. Key assumptions include a cost of equity of 10%, a medium-term growth rate of 15% and a terminal growth rate of 3%.

Consensus Price Target Distribution

Source: Refinitiv, Morgan Stanley Research



RISK REWARD CHART



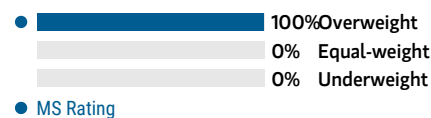
Key: — Historical Stock Performance ● Current Stock Price ◆ Price Target

Source: Refinitiv, Morgan Stanley Research

OVERWEIGHT THESIS

- iPhone volumes are more resilient than Android given Apple's advantages on securing memory.
- ZDT is one of the key ABF substrate suppliers for China AI chips, which is expected to grow meaningfully over the next few years.
- ZDT is growing its ABF exposure to International clients as well, including Google TPU.
- AI optical transceiver module PCB is a meaningful growth driver for ZDT, with very good margins (GM 40%+).
- Shenzhen ABF fab 1 continues to ramp and has turned profitable in 1Q26, and fab 2 is now under construction. Kaohsiung fab is on track to enter mass production in 2H26.
- Our PT implies 27x/20x CY27e/28e P/E.

Consensus Rating Distribution



Source: Refinitiv, Morgan Stanley Research

Risk Reward Themes

Pricing Power: *Negative*

View descriptions of Risk Rewards Themes [here](#)

BULL CASE

NT\$946.00

37x 2027e P/E

Greater share gains and better earnings contribution from new projects in ABF and AI Server PCBs: In the bull case, we assume ZDT is able to do Google TPU substrate extremely well and uses it to approach other ASIC/GPU customers, expanding its international client base. Additionally, we also assume ZDT expands Nvidia AI PCB share to 20-30% share on good execution and yield.

BASE CASE

NT\$666.00

27.5x 2027e P/E

Stable demand for F-PCB and SLP, with share gains in BT/ABF substrates, and AI server PCBs: Zhen Ding is one of the main ABF substrate suppliers for China AI chips, while growing its share for international clients, potentially starting with Google TPU. At the same time, it continues to grow its AI server PCB share with Nvidia and other hyperscalers, and grows its exposure to optical module transceivers.

BEAR CASE

NT\$386.00

16x 2027e P/E

Weaker iPhone demand with share loss and ZDT does not execute on Google TPU substrates: iPhone demand misses market expectations, Zhen Ding loses share in F-PCB to Chinese peers, and SLP demand is also weak. We also assume here that ZDT fails to execute on TPU substrates and only continues to have exposure to China AI chips, with less than 5% share for Nvidia AI PCBs.

Risk Reward – Zhen Ding (4958.TW)

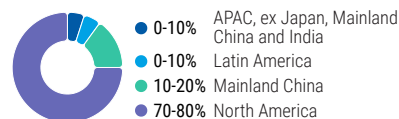
KEY EARNINGS INPUTS

Drivers	2025	2026e	2027e	2028e
F-PCB sales y/y (%)	5.0	1.2	14.6	7.3
F-PCB GM (%)	21.3	24.2	24.6	25.6
F-PCB ASP per sq ft (NT\$)	59.5	60.4	63.1	65.3

INVESTMENT DRIVERS

- iPhone, MacBook, iPad demand
- F-PCB content per iPhone, MacBook, iPad
- F-PCB ASP
- Yield rate for ABF/BT substrates

GLOBAL REVENUE EXPOSURE



Source: Morgan Stanley Research Estimate
View explanation of regional hierarchies [here](#)

MS ALPHA MODELS

3/5
MOST

3 Month
Horizon

Source: Refinitiv, FactSet, Morgan Stanley Research; 1 is the highest favored Quintile and 5 is the least favored Quintile

RISKS TO PT/RATING

RISKS TO UPSIDE

- Higher-than-expected F-PCB content increase for iPhones
- Better-than-expected production yield/share allocation for SLP
- Share gains on higher margin F-PCB pieces

RISKS TO DOWNSIDE

- Worse-than-expected iPhone sell-through
- Lower-than-expected F-PCB content increase for iPhones
- Worse-than-expected production yield/share allocation for SLP
- Increasing competition from Chinese peers, intensifying pricing pressure

OWNERSHIP POSITIONING

Inst. Owners, % Active 65.6%

Source: Refinitiv, Morgan Stanley Research

MS ESTIMATES VS. CONSENSUS

FY Dec 2027e

Sales / Revenue (NT\$, mn)

255,610	291,591	321,782
	284,811	

EBITDA (NT\$, mn)

51,041	70,537	86,189
--------	--------	--------

Net income (NT\$, mn)

18,699	25,909	31,958
	25,067	

EPS (NT\$)

17.47	24.21	29.81
	23.18	

◆ Mean ◆ Morgan Stanley Estimates
Source: Refinitiv, Morgan Stanley Research

Risk Reward – Unimicron (3037.TW)

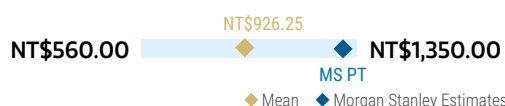
Price hikes and spec upgrades to support higher earnings estimates; OW

PRICE TARGET NT\$1,285.00

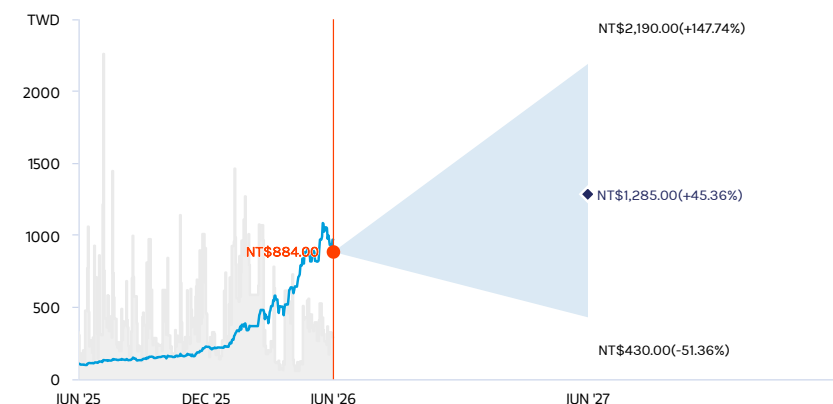
Base case, residual income (RI) valuation model, which we think derives the most accurate value of the firm given that it takes into account cost of equity. We use a cost of equity of 9.2% [risk-free rate of 1% (10-year Taiwan government note yield), equity risk premium of 8.7%, and a beta of 1.0], a medium-term growth rate of 15%, and a terminal growth rate of 3%.

Consensus Price Target Distribution

Source: Refinitiv, Morgan Stanley Research



RISK REWARD CHART



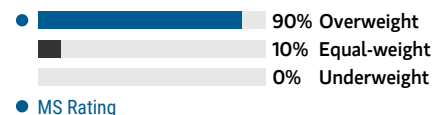
Key: — Historical Stock Performance ● Current Stock Price ◆ Price Target

Source: Refinitiv, Morgan Stanley Research

OVERWEIGHT THESIS

- We expect the ABF substrate market to revert to undersupply from 2027 onwards, with the undersupply gap reaching 5-10% by 2030.
- This is primarily driven by AI demand, with next-gen AI/networking chips adopting larger body size ABF substrates.
- In the near to medium term, the short supply of T-glass will be a key watch point, but we believe this will mainly affect low-end ABF and BT substrates, as suppliers prioritize supply to the AI chip vendors.
- Our price target implies 51x 2027e P/E or 25x 2028e P/E, above the 10-20x range at which it traded in the previous upcycle. We deem this fair, given AI chip substrate spec upgrades and suppliers' ability to raise prices.

Consensus Rating Distribution



Source: Refinitiv, Morgan Stanley Research

Risk Reward Themes

Pricing Power: *Positive*
Secular Growth: *Positive*

View descriptions of Risk Rewards Themes [here](#)

BULL CASE NT\$2,190.00

87x 2027e P/E

Demand for AI servers, general servers and PCs is much stronger than expected: Pricing increases more than expected, 30%+ in 2026-27, driving stronger margin expansion for Unimicron.

BASE CASE NT\$1,285.00

51x 2027e P/E

Entering up-cycle driven by AI chip upgrades: Robust demand for AI GPUs and ASICs, supported by continued data center infrastructure investments by CSPs and neo cloud, leads to undersupply in the high-end ABF market from 2027e onwards. PC demand is sluggish, while general servers still show double-digit unit growth in 2026e. ASP has bottomed and started to inflect upwards, driven by raw material price hikes and substrate spec upgrades.

BEAR CASE NT\$430.00

17x 2027e P/E

Demand for AI servers and general servers weakens, with PC demand weakening further: The result is greater ABF oversupply; thus, there are pricing declines of ~15-20% in 2026-27, leading to weaker margins for Unimicron over the next two years.

Risk Reward – Unimicron (3037.TW)

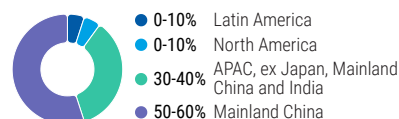
KEY EARNINGS INPUTS

Drivers	2025	2026e	2027e	2028e
IC substrate ASP YoY (%)	7.5	29.7	33.5	29.3
IC substrate GM (%)	15.2	22.3	28.6	37.1

INVESTMENT DRIVERS

- 5G smartphone demand
- 5G base station demand
- Automotive demand

GLOBAL REVENUE EXPOSURE



Source: Morgan Stanley Research Estimate
View explanation of regional hierarchies [here](#)

MS ALPHA MODELS

4/5
MOST
3 Month
Horizon

Source: Refinitiv, FactSet, Morgan Stanley Research; 1 is the highest favored Quintile and 5 is the least favored Quintile

RISKS TO PT/RATING

RISKS TO UPSIDE

- Better-than-expected ABF substrate demand from PC and server customers
- Capex cuts; halt to its capacity expansion plan
- Continued yield issues of alternative technology that doesn't require substrate (e.g., CoWoP)

RISKS TO DOWNSIDE

- Sudden demand shortfall
- Technological change that would not require ABF substrates
- Intensifying competition
- Yield issues or production hiccups when ramping new capacity

OWNERSHIP POSITIONING

Inst. Owners, % Active 68.8%

Source: Refinitiv, Morgan Stanley Research

MS ESTIMATES VS. CONSENSUS

FY Dec 2027e

Sales / Revenue (NT\$, mn)

192,100	262,464	320,188
	241,969	

Net income (NT\$, mn)

28,608	38,875	64,392
	43,767	

EPS (NT\$)

15.89	25.26	41.37
	27.56	

◆ Mean ◆ Morgan Stanley Estimates

Source: Refinitiv, Morgan Stanley Research

Risk Reward – Shennan Circuits Co Ltd (002916.SZ)

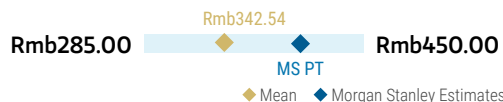
Beneficiary of supply chain localization; EW on valuation

PRICE TARGET Rmb400.00

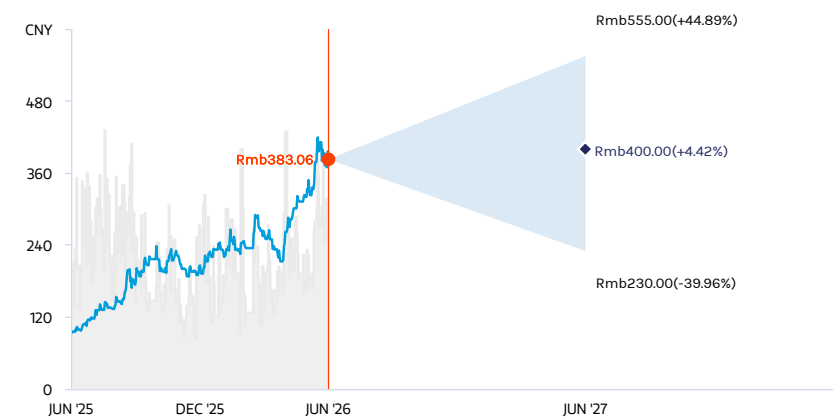
Base case, derived from a residual income (RI) model. Key assumptions include a cost of equity of 6.7% (beta of 1.1, equity premium of 5%, and risk-free rate of 1.3%), 7% medium-term growth rate, and 4% terminal growth rate.

Consensus Price Target Distribution

Source: Refinitiv, Morgan Stanley Research



RISK REWARD CHART



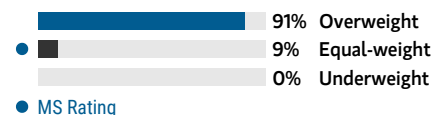
Key: — Historical Stock Performance ● Current Stock Price ◆ Price Target

Source: Refinitiv, Morgan Stanley Research

EQUAL-WEIGHT THESIS

- We view the stock as fairly valued based on current demand indicators, with expectations for rising automotive and data center PCB already elevated for 2026-27.
- Shennan's "China localization" story should continue to play out in the longer term for PCB and IC substrates, and it should be the main beneficiary amid China's semiconductor self-sufficiency drive.
- Demand for 5G base station PCB was slow in 2025, and has remained slow in 2026, with limited visibility on the trajectory of demand improvement (both domestic and overseas).
- Server/Optical transceiver PCB demand is improving for cloud in China, and new platform penetration rate continues to rise.

Consensus Rating Distribution



Source: Refinitiv, Morgan Stanley Research

Risk Reward Themes

Electric Vehicles: *Positive*
 New Data Era: *Positive*
 Pricing Power: *Negative*

View descriptions of Risk Rewards Themes [here](#)

BULL CASE

Rmb555.00

50x 2027e EPS

Shennan's share of the 5G base station market expands to 20-25% (vs. base case scenario of 15-20%), while its automotive and server PCB business doubles over the next two years and its IC substrate business grows 40-50% YoY in 2026 (vs. base case scenario of ~30%). This results in a 55-65% revenue CAGR in 2025-28e.

BASE CASE

Rmb400.00

36x 2027e EPS

We expect Shennan's share of the 5G base station market to be stable at 15-20%, while its server and automotive PCB continues to rise steadily and its IC substrate business continues to outgrow the market in 2026e as they gain share, especially in the domestic market. This results in a 30-35% revenue CAGR in 2025-28e.

BEAR CASE

Rmb230.00

21x 2027e EPS

Shennan's share of the 5G base station market falls to 5-10% (vs. base case scenario of 15-20%), while its server and automotive PCB fails to continue to grow and its IC substrate business grows only slightly above the market in 2026. This results in a 0-15% revenue CAGR in 2025-28e.

Risk Reward – Shennan Circuits Co Ltd (002916.SZ)

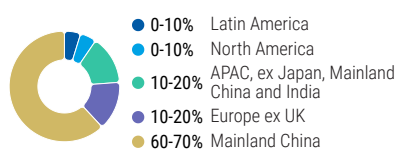
KEY EARNINGS INPUTS

Drivers	2025	2026e	2027e	2028e
PCB revenue y/y (%)	37	38	31	31
IC substrate revenue y/y (%)	31	69	36	27
PCB GM (%)	35.5	34.7	37.0	37.8
IC substrate GM (%)	22.6	30.7	33.6	33.6

INVESTMENT DRIVERS

- 5G base station deployment schedule
- Supplier share allocation
- China's data center demand
- China's semiconductor localization schedule

GLOBAL REVENUE EXPOSURE



Source: Morgan Stanley Research Estimate
View explanation of regional hierarchies [here](#)

MS ALPHA MODELS

3/5
MOST

3 Month
Horizon

Source: Refinitiv, FactSet, Morgan Stanley Research; 1 is the highest favored Quintile and 5 is the least favored Quintile

RISKS TO PT/RATING

RISKS TO UPSIDE

- Rising demand for 5G and data center PCB.
- Further share gain for 5G and data center PCB.
- Rising IC substrate demand driven by acceleration in China's semiconductor localization.

RISKS TO DOWNSIDE

- Rising competition for 5G and data center PCB.
- Delay in 5G network deployment.
- Escalating US-China trade tensions.

OWNERSHIP POSITIONING

Inst. Owners, % Active **85.3%**

Source: Refinitiv, Morgan Stanley Research

MS ESTIMATES VS. CONSENSUS

FY Dec 2027e

Sales / Revenue (Rmb, mn) 33,982.4 **42,153.6** 39,067.8 43,360.8

EBIT (Rmb, mn) 6,794.7 **8,717** 7,724.5 8,952.3

Net income (Rmb, mn) 6,163.2 **7,493** 7,194.9 8,477.0

EPS (Rmb) 9.2 **11.2** 10.6 12.4

ROE (%) 27.3 **36.1** 29.2

◆ Mean ◆ Morgan Stanley Estimates

Source: Refinitiv, Morgan Stanley Research

Risk Reward Reference links

1. View explanation of Options Probabilities methodology - [Options_Probabilities_Exhibit_Link.pdf](#)
2. View descriptions of Risk Rewards Themes - [RR_Themes_Exhibit_Link.pdf](#)
3. View explanation of regional hierarchies - [GEG_Exhibit_Link.pdf](#)
4. View explanation of Theme/Exposure methodology - [ESG_Sustainable_Solutions_External_Link.pdf](#)
5. View explanation of HERS methodology - [ESG_HERS_External_Link.pdf](#)

Disclosure Section

The information and opinions in Morgan Stanley Research were prepared or are disseminated by Morgan Stanley Asia Limited (which accepts the responsibility for its contents) and/or Morgan Stanley Asia (Singapore) Pte. (Registration number 199206298Z) and/or Morgan Stanley Asia (Singapore) Securities Pte Ltd (Registration number 200008434H), regulated by the Monetary Authority of Singapore (which accepts legal responsibility for its contents and should be contacted with respect to any matters arising from, or in connection with, Morgan Stanley Research), and/or Morgan Stanley Taiwan Limited and/or Morgan Stanley & Co International plc, Seoul Branch, and/or Morgan Stanley Australia Limited (A.B.N. 67 003 734 576, holder of Australian financial services license No. 233742, which accepts responsibility for its contents), and/or Morgan Stanley Wealth Management Australia Pty Ltd (A.B.N. 19 009 145 555, holder of Australian financial services license No. 240813, which accepts responsibility for its contents), and/or Morgan Stanley India Company Private Limited having Corporate Identification No (CIN) U22990MH1998PTC115305, regulated by the Securities and Exchange Board of India ("SEBI") and holder of licenses as a Research Analyst (SEBI Registration No. INH000001105); Stock Broker (SEBI Stock Broker Registration No. INZ000244438), Merchant Banker (SEBI Registration No. INM000011203), and depository participant with National Securities Depository Limited (SEBI Registration No. IN-DP-NSDL-567-2021) having registered office at Altimus, Level 39 & 40, Pandurang Budhkar Marg, Worli, Mumbai 400018, India; Telephone no. +91-22-61181000; Compliance Officer Details: Mr. Tejarshi Hardas, Tel. No.: +91-22-61181000 or Email: tejarshi.hardas@morganstanley.com; Grievance officer details: Mr. Tejarshi Hardas, Tel. No.: +91-22-61181000 or Email: msic-compliance@morganstanley.com which accepts the responsibility for its contents and should be contacted with respect to any matters arising from, or in connection with, Morgan Stanley Research, and their affiliates (collectively, "Morgan Stanley"). Morgan Stanley India Company Private Limited (MSICPL) may use AI tools in providing research services. All recommendations contained herein are made by the duly qualified research analysts.

For important disclosures, stock price charts and equity rating histories regarding companies that are the subject of this report, please see the Morgan Stanley Research Disclosure Website at www.morganstanley.com/eqr/disclosures/webapp/generalresearch, or contact your investment representative or Morgan Stanley Research at 1585 Broadway, (Attention: Research Management), New York, NY, 10036 USA.

For valuation methodology and risks associated with any recommendation, rating or price target referenced in this research report, please contact the Client Support Team as follows: US/Canada +1 800 303-2495; Hong Kong +852 2848-5999; Latin America +1 718 754-5444 (U.S.); London +44 (0)20-7425-8169; Singapore +65 6834-6860; Sydney +61 (0)2-9770-1505; Tokyo +81 (0)3-6836-9000. Alternatively you may contact your investment representative or Morgan Stanley Research at 1585 Broadway, (Attention: Research Management), New York, NY 10036 USA.

Analyst Certification

The following analysts hereby certify that their views about the companies and their securities discussed in this report are accurately expressed and that they have not received and will not receive direct or indirect compensation in exchange for expressing specific recommendations or views in this report: Howard Kao; Sharon Shih.

Global Research Conflict Management Policy

Morgan Stanley Research has been published in accordance with our conflict management policy, which is available at www.morganstanley.com/institutional/research/conflictolicies. A Portuguese version of the policy can be found at www.morganstanley.com.br

Important Regulatory Disclosures on Subject Companies

As of May 29, 2026, Morgan Stanley beneficially owned 1% or more of a class of common equity securities of the following companies covered in Morgan Stanley Research: AAC Technologies Holdings, Accelink Technologies Co. Ltd., Acer Inc., AirTAC International, Asia Vital Components Co. Ltd., AU Optronics, Auras Technology Co Ltd, Bizlink, Catcher Technology, Chenbro, Compal Electronics, Delta Electronics Inc., E Ink Holdings Inc., Eoptolink Technology Inc Ltd, Fositek Corp, Genius Electronic Optical Co. Ltd., Giga-Byte Technology Co. Ltd., Gold Circuit Electronics Ltd., Hiwin Technologies Corp., Innolux, LandMark Optoelectronics Corporation, Lite-On Technology, Nan Ya PCB, Pegatron Corporation, Sunny Optical, Sunonwealth Electric Machine Industry Co, Suzhou TFC Optical Communication Co Ltd., TCL Corp., Tong Hsing, Unimicron, Visual Photonics Epitaxy Co Ltd, Wistron Corporation, Wiwynn Corp, Wuhan Jingce Electronic Group Co Ltd, Xiaomi Corp, Yageo Corp., Zhejiang Crystal-Optech Co Ltd, Zhen Ding, Zhongji Innolight Co Ltd, ZTE Corporation.

Within the last 12 months, Morgan Stanley managed or co-managed a public offering (or 144A offering) of securities of Wistron Corporation, Wiwynn Corp, Zhen Ding.

Within the last 12 months, Morgan Stanley has received compensation for investment banking services from Lenovo, Wistron Corporation, Wiwynn Corp, Xiaomi Corp, Yageo Corp., Zhen Ding. In the next 3 months, Morgan Stanley expects to receive or intends to seek compensation for investment banking services from AAC Technologies Holdings, Accton Technology Corporation, Acer Inc., Advantech, AirTAC International, Asia Vital Components Co. Ltd., Asustek Computer Inc., AU Optronics, Bizlink, Catcher Technology, Chenbro, Compal Electronics, Delta Electronics Inc., E Ink Holdings Inc., Ennostar Inc, Eoptolink Technology Inc Ltd, FIT Hon Teng Ltd, Giga-Byte Technology Co. Ltd., GoerTek Inc, Gold Circuit Electronics Ltd., Hon Hai Precision, Innolux, Lenovo, Lens Technology, Lingyi Itech Guangdong Co, Lite-On Technology, Luxshare Precision Industry Co., Ltd., Pegatron Corporation, Q Technology (Group) Company Ltd, Quanta Computer Inc., Sanan Optoelectronics, Shenzhen Transsion Holdings Co Ltd, Suzhou TFC Optical Communication Co Ltd., TCL Corp., Unimicron, Wistron Corporation, Wiwynn Corp, Xiaomi Corp, Yageo Corp., Yangtze Optical Fibre and Cable JSC Ltd, Zhen Ding, Zhongji Innolight Co Ltd.

Within the last 12 months, Morgan Stanley has received compensation for products and services other than investment banking services from AAC Technologies Holdings, Accton Technology Corporation, Acer Inc., Asustek Computer Inc., AU Optronics, BYD Electronics, Compal Electronics, E Ink Holdings Inc., Eoptolink Technology Inc Ltd, Foxconn Technology, Giga-Byte Technology Co. Ltd., GoerTek Inc, Hon Hai Precision, Innolux, Lenovo, Lingyi Itech Guangdong Co, Quanta Computer Inc., Xiaomi Corp, Yageo Corp..

Within the last 12 months, Morgan Stanley has provided or is providing investment banking services to, or has an investment banking client relationship with, the following company: AAC Technologies Holdings, Accton Technology Corporation, Acer Inc., Advantech, AirTAC International, Asia Vital Components Co. Ltd., Asustek Computer Inc., AU Optronics, Bizlink, Catcher Technology, Chenbro, Compal Electronics, Delta Electronics Inc., E Ink Holdings Inc., Ennostar Inc, Eoptolink Technology Inc Ltd, FIT Hon Teng Ltd, Giga-Byte Technology Co. Ltd., GoerTek Inc, Gold Circuit Electronics Ltd., Hon Hai Precision, Innolux, Lenovo, Lens Technology, Lingyi Itech Guangdong Co, Lite-On Technology, Luxshare Precision Industry Co., Ltd., Pegatron Corporation, Q Technology (Group) Company Ltd, Quanta Computer Inc., Sanan Optoelectronics, Shenzhen Transsion Holdings Co Ltd, Suzhou TFC Optical Communication Co Ltd., TCL Corp., Unimicron, Wistron Corporation, Wiwynn Corp, Xiaomi Corp, Yageo Corp., Yangtze Optical Fibre and Cable JSC Ltd, Zhen Ding, Zhongji Innolight Co Ltd.

Within the last 12 months, Morgan Stanley has either provided or is providing non-investment banking, securities-related services to and/or in the past has entered into an agreement to provide services or has a client relationship with the following company: AAC Technologies Holdings, Accton Technology Corporation, Acer Inc., Asustek Computer Inc., AU Optronics, BYD Electronics, Compal Electronics, E Ink Holdings Inc., Eoptolink Technology Inc Ltd, Foxconn Technology, Giga-Byte Technology Co. Ltd., GoerTek Inc, Hon Hai Precision, Innolux, Lenovo, Lingyi Itech Guangdong Co, Quanta Computer Inc., Xiaomi Corp, Yageo Corp., Zhen Ding.

The equity research analysts or strategists principally responsible for the preparation of Morgan Stanley Research have received compensation based upon various factors, including quality of research, investor client feedback, stock picking, competitive factors, firm revenues and overall investment banking revenues. Equity Research analysts' or strategists' compensation is not linked to investment banking or capital markets transactions performed by Morgan Stanley or the profitability or revenues of particular trading desks.

Morgan Stanley and its affiliates do business that relates to companies/instruments covered in Morgan Stanley Research, including market making, providing liquidity, fund management,

commercial banking, extension of credit, investment services and investment banking. Morgan Stanley sells to and buys from customers the securities/instruments of companies covered in Morgan Stanley Research on a principal basis. Morgan Stanley may have a position in the debt of the Company or instruments discussed in this report. Morgan Stanley trades or may trade as principal in the debt securities (or in related derivatives) that are the subject of the debt research report.

Certain disclosures listed above are also for compliance with applicable regulations in non-US jurisdictions.

STOCK RATINGS

Morgan Stanley uses a relative rating system using terms such as Overweight, Equal-weight, Not-Rated or Underweight (see definitions below). Morgan Stanley does not assign ratings of Buy, Hold or Sell to the stocks we cover. Overweight, Equal-weight, Not-Rated and Underweight are not the equivalent of buy, hold and sell. Investors should carefully read the definitions of all ratings used in Morgan Stanley Research. In addition, since Morgan Stanley Research contains more complete information concerning the analyst's views, investors should carefully read Morgan Stanley Research, in its entirety, and not infer the contents from the rating alone. In any case, ratings (or research) should not be used or relied upon as investment advice. An investor's decision to buy or sell a stock should depend on individual circumstances (such as the investor's existing holdings) and other considerations.

Global Stock Ratings Distribution

(as of May 31, 2026)

The Stock Ratings described below apply to Morgan Stanley's Fundamental Equity Research and do not apply to Debt Research produced by the Firm.

For disclosure purposes only (in accordance with FINRA requirements), we include the category headings of Buy, Hold, and Sell alongside our ratings of Overweight, Equal-weight, Not-Rated and Underweight. Morgan Stanley does not assign ratings of Buy, Hold or Sell to the stocks we cover. Overweight, Equal-weight, Not-Rated and Underweight are not the equivalent of buy, hold, and sell but represent recommended relative weightings (see definitions below). To satisfy regulatory requirements, we correspond Overweight, our most positive stock rating, with a buy recommendation; we correspond Equal-weight and Not-Rated to hold and Underweight to sell recommendations, respectively.

Stock Rating Category	Coverage Universe		Investment Banking Clients (IBC)			Other Material Investment Services Clients (MISC)	
	Count	% of Total	Count	% of Total IBC	% of Rating Category	Count	% of Total Other MISC
Overweight/Buy	1542	42%	465	51%	30%	707	43%
Equal-weight/Hold	1571	43%	369	40%	23%	723	44%
Not-Rated/Hold	3	0%	0	0%	0%	1	0%
Underweight/Sell	551	15%	86	9%	16%	201	12%
Total	3,667		920			1632	

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

Analyst Stock Ratings

Overweight (O). The stock's total return is expected to exceed the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Equal-weight (E). The stock's total return is expected to be in line with the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Not-Rated (NR). Currently the analyst does not have adequate conviction about the stock's total return relative to the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Underweight (U). The stock's total return is expected to be below the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Unless otherwise specified, the time frame for price targets included in Morgan Stanley Research is 12 to 18 months.

Analyst Industry Views

Attractive (A): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be attractive vs. the relevant broad market benchmark, as indicated below.

In-Line (I): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be in line with the relevant broad market benchmark, as indicated below.

Cautious (C): The analyst views the performance of his or her industry coverage universe over the next 12-18 months with caution vs. the relevant broad market benchmark, as indicated below.

Benchmarks for each region are as follows: North America - S&P 500; Latin America - relevant MSCI country index or MSCI Latin America Index; Europe - MSCI Europe; Japan - TOPIX; Asia - relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

Stock Price, Price Target and Rating History (See Rating Definitions)

Shennan Circuits Co Ltd (002916.SZ) - As of 06/11/26 GMT in CNY
Industry : Greater China Technology Hardware



Stock Rating History: 6/1/21 : O/I; 8/24/23 : E/I

Price Target History: 10/29/20 : 111.54; 6/4/21 : 96.15; 8/22/22 : 88.46; 10/31/22 : 76.92; 12/21/22 : 69.23; 8/24/23 : 53.85; 10/30/23 : 55.38; 3/17/24 : 63.08; 9/15/24 : 79.23; 10/31/24 : 80; 10/31/25 : 235; 3/16/26 : 260; 4/20/26 : 280; 4/27/26 : 320

Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target --- No Price Target Assigned (NA)

Stock Price (Not Covered by Current Analyst) — Stock Price (Covered by Current Analyst) —

Stock and Industry Ratings (abbreviations below) appear as ♦ Stock Rating/Industry View

Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)

Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

Effective January 13, 2014, the stocks covered by Morgan Stanley Asia Pacific will be rated relative to the analyst's industry (or industry team's) coverage.

Effective January 13, 2014, the industry view benchmarks for Morgan Stanley Asia Pacific are as follows: relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

Unimicron (3037.TW) - As of 06/11/26 GMT in TWD
Industry : Greater China Technology Hardware



Stock Rating History: 6/1/21 : NA/I; 10/22/21 : O/I; 6/1/22 : E/I; 2/22/23 : U/I; 4/15/25 : E/I; 2/23/26 : O/I

Price Target History: 10/22/21 : 188.06; 10/27/21 : 237.55; 2/3/22 : 267.24; 2/24/22 : 366.22; 6/1/22 : 237.55; 8/23/22 : 178.16; 10/27/22 : 133.62; 12/21/22 : 123.72; 2/2/23 : 116.79; 2/22/23 : 98.98; 6/7/23 : 138.57; 11/7/23 : 125.7; 1/24/24 : 118.77; 4/15/25 : 81.66; 7/30/25 : 120.75; 2/23/26 : 500; 2/26/26 : 550; 4/20/26 : 785; 5/18/26 : 1225

Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target --- No Price Target Assigned (NA)

Stock Price (Not Covered by Current Analyst) — Stock Price (Covered by Current Analyst) —

Stock and Industry Ratings (abbreviations below) appear as ♦ Stock Rating/Industry View

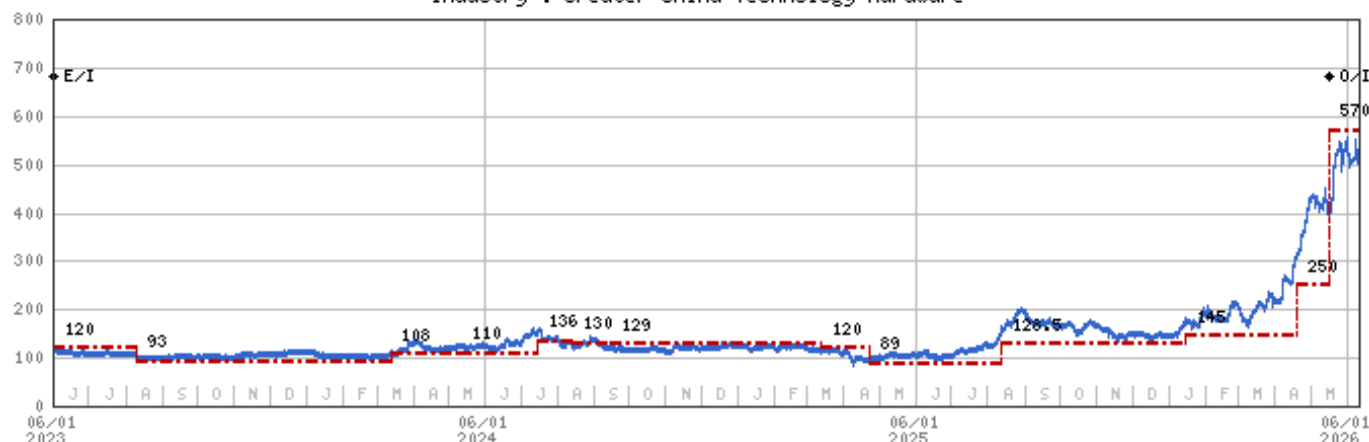
Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)

Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

Effective January 13, 2014, the stocks covered by Morgan Stanley Asia Pacific will be rated relative to the analyst's industry (or industry team's) coverage.

Effective January 13, 2014, the industry view benchmarks for Morgan Stanley Asia Pacific are as follows: relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

Zhen Ding (4958.TW) - As of 06/11/26 GMT in TWD
Industry : Greater China Technology Hardware



Stock Rating History: 6/1/21 : U/I; 8/2/22 : E/I; 5/18/26 : O/I

Price Target History: 3/30/21 : 105; 8/2/22 : 124; 11/4/22 : 126; 3/14/23 : 120; 8/10/23 : 93; 3/13/24 : 108; 5/11/24 : 110; 7/15/24 : 136; 8/13/24 : 130; 9/15/24 : 129; 3/12/25 : 120; 4/23/25 : 89; 8/13/25 : 128.5; 1/16/26 : 145; 4/20/26 : 250; 5/18/26 : 570

Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target --- No Price Target Assigned (NA)
Stock Price (Not Covered by Current Analyst) — Stock Price (Covered by Current Analyst) —
Stock and Industry Ratings (abbreviations below) appear as ♦ Stock Rating/Industry View
Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)
Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

Effective January 13, 2014, the stocks covered by Morgan Stanley Asia Pacific will be rated relative to the analyst's industry (or industry team's) coverage.

Effective January 13, 2014, the industry view benchmarks for Morgan Stanley Asia Pacific are as follows: relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

Important Disclosures for Morgan Stanley Smith Barney LLC Customers

Important disclosures regarding any material conflict of interest that can reasonably be expected to have influenced Morgan Stanley Smith Barney LLC's choice of a third-party research provider or the subject company of a third-party research report, are available on the Morgan Stanley Wealth Management disclosure website at www.morganstanley.com/online/researchdisclosures. For Morgan Stanley specific disclosures, you may refer to <https://www.morganstanley.com/eqr/disclosures/webapp/generalresearch>.

Each Morgan Stanley research report is reviewed and approved on behalf of Morgan Stanley Smith Barney LLC. This review and approval is conducted by the same person who reviews the research report on behalf of Morgan Stanley. This could create a conflict of interest.

Other Important Disclosures

Morgan Stanley Research policy is to update research reports as and when the Research Analyst and Research Management deem appropriate, based on developments with the issuer, the sector, or the market that may have a material impact on the research views or opinions stated therein. In addition, certain Research publications are intended to be updated on a regular periodic basis (weekly/monthly/quarterly/annual) and will ordinarily be updated with that frequency, unless the Research Analyst and Research Management determine that a different publication schedule is appropriate based on current conditions.

Morgan Stanley is not acting as a municipal advisor and the opinions or views contained herein are not intended to be, and do not constitute, advice within the meaning of Section 975 of the Dodd-Frank Wall Street Reform and Consumer Protection Act.

Morgan Stanley produces an equity research product called a "Tactical Idea." Views contained in a "Tactical Idea" on a particular stock may be contrary to the recommendations or views expressed in research on the same stock. This may be the result of differing time horizons, methodologies, market events, or other factors. For all research available on a particular stock, please contact your sales representative or go to Matrix at <http://www.morganstanley.com/matrix>.

Morgan Stanley Research is provided to our clients through our proprietary research portal on Matrix and also distributed electronically by Morgan Stanley to clients. Certain, but not all, Morgan Stanley Research products are also made available to clients through third-party vendors or redistributed to clients through alternate electronic means as a convenience. For access to all available Morgan Stanley Research, please contact your sales representative or go to Matrix at <http://www.morganstanley.com/matrix>.

Any access and/or use of Morgan Stanley Research is subject to Morgan Stanley's Terms of Use (<http://www.morganstanley.com/terms.html>). By accessing and/or using Morgan Stanley Research, you are indicating that you have read and agree to be bound by our Terms of Use (<http://www.morganstanley.com/terms.html>). In addition you consent to Morgan Stanley processing your personal data and using cookies in accordance with our Privacy Policy and our Global Cookies Policy (http://www.morganstanley.com/privacy_pledge.html), including for the purposes of setting your preferences and to collect readership data so that we can deliver better and more personalized service and products to you. To find out more information about how Morgan Stanley processes personal data, how we use cookies and how to reject cookies see our Privacy Policy and our Global Cookies Policy (http://www.morganstanley.com/privacy_pledge.html). Please use the provided link to review the Terms and Conditions and Most Important Terms and Conditions for Morgan Stanley India Company Private Limited (https://www.morganstanley.com/assets/pdfs/about-us-global-offices/india/Terms_and_conditions.pdf) and the following link to review the audit report (https://ny.matrix.ms.com/eqr/research/webapp/researchdocs/MSICPL_Morgan_Stanley_Research_Audit_Report.pdf).

If you do not agree to our Terms of Use and/or if you do not wish to provide your consent to Morgan Stanley processing your personal data or using cookies please do not access our research. Morgan Stanley Research does not provide individually tailored investment advice. Morgan Stanley Research has been prepared without regard to the circumstances and objectives of those

who receive it. Morgan Stanley recommends that investors independently evaluate particular investments and strategies, and encourages investors to seek the advice of a financial adviser. The appropriateness of an investment or strategy will depend on an investor's circumstances and objectives. The securities, instruments, or strategies discussed in Morgan Stanley Research may not be suitable for all investors, and certain investors may not be eligible to purchase or participate in some or all of them. Morgan Stanley Research is not an offer to buy or sell or the solicitation of an offer to buy or sell any security/instrument or to participate in any particular trading strategy. The value of and income from your investments may vary because of changes in interest rates, foreign exchange rates, default rates, prepayment rates, securities/instruments prices, market indexes, operational or financial conditions of companies or other factors. There may be time limitations on the exercise of options or other rights in securities/instruments transactions. Past performance is not necessarily a guide to future performance. Estimates of future performance are based on assumptions that may not be realized. If provided, and unless otherwise stated, the closing price on the cover page is that of the primary exchange for the subject company's securities/instruments.

The fixed income research analysts, strategists or economists principally responsible for the preparation of Morgan Stanley Research have received compensation based upon various factors, including quality, accuracy and value of research, firm profitability or revenues (which include fixed income trading and capital markets profitability or revenues), client feedback and competitive factors. Fixed Income Research analysts', strategists' or economists' compensation is not linked to investment banking or capital markets transactions performed by Morgan Stanley or the profitability or revenues of particular trading desks.

The "Important Regulatory Disclosures on Subject Companies" section in Morgan Stanley Research lists all companies mentioned where Morgan Stanley owns 1% or more of a class of common equity securities of the companies. For all other companies mentioned in Morgan Stanley Research, Morgan Stanley may have an investment of less than 1% in securities/instruments or derivatives of securities/instruments of companies and may trade them in ways different from those discussed in Morgan Stanley Research. Employees of Morgan Stanley not involved in the preparation of Morgan Stanley Research may have investments in securities/instruments or derivatives of securities/instruments of companies mentioned and may trade them in ways different from those discussed in Morgan Stanley Research. Derivatives may be issued by Morgan Stanley or associated persons.

With the exception of information regarding Morgan Stanley, Morgan Stanley Research is based on public information. Morgan Stanley makes every effort to use reliable, comprehensive information, but we make no representation that it is accurate or complete. We have no obligation to tell you when opinions or information in Morgan Stanley Research change apart from when we intend to discontinue equity research coverage of a subject company. Facts and views presented in Morgan Stanley Research have not been reviewed by, and may not reflect information known to, professionals in other Morgan Stanley business areas, including investment banking personnel.

Morgan Stanley Research personnel may participate in company events such as site visits and are generally prohibited from accepting payment by the company of associated expenses unless pre-approved by authorized members of Research management.

Morgan Stanley may make investment decisions that are inconsistent with the recommendations or views in this report.

To our readers based in Taiwan or trading in Taiwan securities/instruments: Information on securities/instruments that trade in Taiwan is distributed by Morgan Stanley Taiwan Limited ("MSTL"). Such information is for your reference only. The reader should independently evaluate the investment risks and is solely responsible for their investment decisions. Morgan Stanley Research may not be distributed to the public media or quoted or used by the public media without the express written consent of Morgan Stanley. Any non-customer reader within the scope of Article 7-1 of the Taiwan Stock Exchange Recommendation Regulations accessing and/or receiving Morgan Stanley Research is not permitted to provide Morgan Stanley Research to any third party (including but not limited to related parties, affiliated companies and any other third parties) or engage in any activities regarding Morgan Stanley Research which may create or give the appearance of creating a conflict of interest. Information on securities/instruments that do not trade in Taiwan is for informational purposes only and is not to be construed as a recommendation or a solicitation to trade in such securities/instruments. MSTL may not execute transactions for clients in these securities/instruments.

Certain information in Morgan Stanley Research was sourced by employees of the Shanghai Representative Office of Morgan Stanley Asia Limited for the use of Morgan Stanley Asia Limited. Morgan Stanley is not incorporated under PRC law and the research in relation to this report is conducted outside the PRC. Morgan Stanley Research does not constitute an offer to sell or the solicitation of an offer to buy any securities in the PRC. PRC investors shall have the relevant qualifications to invest in such securities and shall be responsible for obtaining all relevant approvals, licenses, verifications and/or registrations from the relevant governmental authorities themselves. Neither this report nor any part of it is intended as, or shall constitute, provision of any consultancy or advisory service of securities investment as defined under PRC law. Such information is provided for your reference only.

Morgan Stanley Research is disseminated in Brazil by Morgan Stanley C.T.V.M. S.A. located at Av. Brigadeiro Faria Lima, 3600, 6th floor, São Paulo - SP, Brazil; and is regulated by the Comissão de Valores Mobiliários; in Mexico by Morgan Stanley México, Casa de Bolsa, S.A. de C.V. which is regulated by Comisión Nacional Bancaria y de Valores. Paseo de los Tamarindos 90, Torre 1, Col. Bosques de las Lomas Floor 29, 05120 Mexico City; in Japan by Morgan Stanley MUFG Securities Co., Ltd. and, for Commodities related research reports only, Morgan Stanley Capital Group Japan Co., Ltd; in Hong Kong by Morgan Stanley Asia Limited (which accepts responsibility for its contents) and by Morgan Stanley Bank Asia Limited; in Singapore by Morgan Stanley Asia (Singapore) Pte. (Registration number 199206298Z) and/or Morgan Stanley Asia (Singapore) Securities Pte Ltd (Registration number 200008434H), regulated by the Monetary Authority of Singapore (which accepts legal responsibility for its contents and should be contacted with respect to any matters arising from, or in connection with, Morgan Stanley Research) and by Morgan Stanley Bank Asia Limited, Singapore Branch (Registration number T14FC0118); in Australia to "wholesale clients" within the meaning of the Australian Corporations Act by Morgan Stanley Australia Limited A.B.N. 67 003 734 576, holder of Australian financial services license No. 233742, which accepts responsibility for its contents; in Australia to "wholesale clients" and "retail clients" within the meaning of the Australian Corporations Act by Morgan Stanley Wealth Management Australia Pty Ltd (A.B.N. 19 009 145 555, holder of Australian financial services license No. 240813, which accepts responsibility for its contents; in Korea by Morgan Stanley & Co International plc, Seoul Branch; in India by Morgan Stanley India Company Private Limited having Corporate Identification No (CIN) U22990MH1998PTC115305, regulated by the Securities and Exchange Board of India ("SEBI") and holder of licenses as a Research Analyst (SEBI Registration No. INH000001105); Stock Broker (SEBI Stock Broker Registration No. INZ000244438), Merchant Banker (SEBI Registration No. INM000011203), and depository participant with National Securities Depository Limited (SEBI Registration No. IN-DP-NSDL-567-2021) having registered office at Altimus, Level 39 & 40, Pandurang Budhkar Marg, Worli, Mumbai 400018, India; Telephone no. +91-22-61181000; Compliance Officer Details: Mr. Tejarshi Hardas, Tel. No.: +91-22-61181000 or Email: tejarshi.hardas@morganstanley.com; Grievance officer details: Mr. Tejarshi Hardas, Tel. No.: +91-22-61181000 or Email: msic-compliance@morganstanley.com. Morgan Stanley India Company Private Limited (MSICPL) may use AI tools in providing research services. All recommendations contained herein are made by the duly qualified research analysts; in Canada by Morgan Stanley Canada Limited; in Germany and the European Economic Area where required by Morgan Stanley Europe S.E., authorised and regulated by Bundesanstalt fuer Finanzdienstleistungsaufsicht (BaFin) under the reference number 149169; in the US by Morgan Stanley & Co. LLC, which accepts responsibility for its contents. Morgan Stanley & Co. International plc, authorized by the Prudential Regulation Authority and regulated by the Financial Conduct Authority and the Prudential Regulation Authority, disseminates in the UK research that it has prepared, and research which has been prepared by any of its affiliates, only to persons who (i) are investment professionals falling within Article 19(5) of the Financial Services and Markets Act 2000 (Financial Promotion) Order 2005 (as amended, the "Order"); (ii) are persons who are high net worth entities falling within Article 49(2)(a) to (d) of the Order; or (iii) are persons to whom an invitation or inducement to engage in investment activity (within the meaning of section 21 of the Financial Services and Markets Act 2000, as amended) may otherwise lawfully be communicated or caused to be communicated. RMB Morgan Stanley Proprietary Limited is a member of the JSE Limited and A2X (Pty) Ltd. RMB Morgan Stanley Proprietary Limited is a joint venture owned equally by Morgan Stanley International Holdings Inc. and RMB Investment Advisory (Proprietary) Limited, which is wholly owned by FirstRand Limited. The information in Morgan Stanley Research is being disseminated by Morgan Stanley Saudi Arabia, regulated by the Capital Market Authority in the Kingdom of Saudi Arabia, and is directed at Sophisticated investors only.

Morgan Stanley Hong Kong Securities Limited is the liquidity provider/market maker for securities of AAC Technologies Holdings, BYD Electronics, FIT Hon Teng Ltd, Hon Hai Precision, Lenovo, Lens Technology, Sunny Optical, Xiaomi Corp, Yangtze Optical Fibre and Cable JSC Ltd, ZTE Corporation listed on the Stock Exchange of Hong Kong Limited. An updated list can be found on HKEx website: <http://www.hkex.com.hk>.

The information in Morgan Stanley Research is being communicated by Morgan Stanley & Co. International plc (DIFC Branch), regulated by the Dubai Financial Services Authority (the DFSA) or by Morgan Stanley & Co. International plc (ADGM Branch), regulated by the Financial Services Regulatory Authority Abu Dhabi (the FSRA), and is directed at Professional Clients only, as defined by the DFSA or the FSRA, respectively. The financial products or financial services to which this research relates will only be made available to a customer who we are satisfied meets the regulatory criteria of a Professional Client. A distribution of the different MS Research ratings or recommendations, in percentage terms for Investments in each sector covered, is available upon request from your sales representative.

The information in Morgan Stanley Research is being communicated by Morgan Stanley & Co. International plc (QFC Branch), regulated by the Qatar Financial Centre Regulatory Authority (the QFCRA), and is directed at business customers and market counterparties only and is not intended for Retail Customers as defined by the QFCRA.

As required by the Capital Markets Board of Turkey, investment information, comments and recommendations stated here, are not within the scope of investment advisory activity. Investment advisory service is provided exclusively to persons based on their risk and income preferences by the authorized firms. Comments and recommendations stated here are general in nature. These opinions may not fit to your financial status, risk and return preferences. For this reason, to make an investment decision by relying solely to this information stated here may not bring about outcomes that fit your expectations.

The trademarks and service marks contained in Morgan Stanley Research are the property of their respective owners. Third-party data providers make no warranties or representations relating to the accuracy, completeness, or timeliness of the data they provide and shall not have liability for any damages relating to such data. The Global Industry Classification Standard (GICS) was developed by and is the exclusive property of MSCI and S&P.

Morgan Stanley Research, or any portion thereof may not be reprinted, sold or redistributed without the written consent of Morgan Stanley.

Indicators and trackers referenced in Morgan Stanley Research may not be used as, or treated as, a benchmark under Regulation EU 2016/1011, or any other similar framework.

The issuers and/or fixed income products recommended or discussed in certain fixed income research reports may not be continuously followed. Accordingly, investors should regard those fixed income research reports as providing stand-alone analysis and should not expect continuing analysis or additional reports relating to such issuers and/or individual fixed income products.

Morgan Stanley may hold, from time to time, material financial and commercial interests regarding the company subject to the Research report.

Registration granted by SEBI and certification from the National Institute of Securities Markets (NISM) in no way guarantee performance of the intermediary or provide any assurance of returns to investors. Investment in securities market are subject to market risks. Read all the related documents carefully before investing.

INDUSTRY COVERAGE: Greater China Technology Hardware

COMPANY (TICKER)	RATING (AS OF)	PRICE* (06/11/2026)
Andy Meng, CFA		
AAC Technologies Holdings (2018.HK)	O (01/29/2024)	HK\$42.60
Accelink Technologies Co. Ltd. (002281.SZ)	U (05/12/2022)	Rmb205.40
BYD Electronics (0285.HK)	O (04/28/2023)	HK\$24.50
China TransInfo Technology Co Ltd (002373.SZ)	E (07/18/2023)	Rmb7.22
Dahua Technology Co. Ltd. (002236.SZ)	E (12/12/2024)	Rmb15.84
Eoptolink Technology Inc Ltd (300502.SZ)	O (02/27/2026)	Rmb526.00
Genius Electronic Optical Co. Ltd. (3406.TW)	E (04/23/2025)	NT\$693.00
Gosuncn Technology Group Co Ltd (300098.SZ)	U (11/07/2022)	Rmb4.90
HIKVision Digital Technology (002415.SZ)	E (12/12/2024)	Rmb29.98
Largan Precision (3008.TW)	E (10/17/2025)	NT\$4,165.00
LianChuang Electronic Technology Co Ltd (002036.SZ)	U (06/12/2024)	Rmb7.96
OFILM Group Co Ltd (002456.SZ)	U (06/12/2024)	Rmb8.67
Q Technology (Group) Company Ltd (1478.HK)	U (03/23/2026)	HK\$7.98
Quectel Wireless Solutions Co Ltd (603236.SS)	E (10/09/2024)	Rmb56.15
Shanghai Conant Optical Co Ltd (2276.HK)	O (04/10/2026)	HK\$38.40
Shenzhen Transsion Holdings Co Ltd (688036.SS)	E (03/23/2026)	Rmb53.17
Sunny Optical (2382.HK)	E (03/23/2026)	HK\$70.95
Suzhou TFC Optical Communication Co Ltd. (300394.SZ)	E (11/06/2025)	Rmb412.30
Wingtech Technology Co Ltd (600745.SS)	U (05/18/2026)	Rmb17.94
Xiaomi Corp (1810.HK)	O (04/14/2021)	HK\$25.84
Yangtze Optical Fibre and Cable JSC Ltd (601869.SS)	U (10/13/2021)	Rmb442.51
Yangtze Optical Fibre and Cable JSC Ltd (6869.HK)	E (04/20/2023)	HK\$230.60
Yongxin Optics Co Ltd (603297.SS)	E (11/15/2022)	Rmb117.96
Zhejiang Crystal-Optech Co Ltd (002273.SZ)	O (11/15/2022)	Rmb33.18
Zhongji Innolight Co Ltd (300308.SZ)	++	Rmb1,124.00
ZTE Corporation (0763.HK)	O (06/04/2026)	HK\$26.20

ZTE Corporation (000063.SZ)	E (06/04/2026)	Rmb37.81
Derrick Yang		
Accton Technology Corporation (2345.TW)	O (06/06/2024)	NT\$2,275.00
Advantech (2395.TW)	O (01/20/2021)	NT\$464.00
AirTAC International (1590.TW)	O (04/16/2025)	NT\$1,265.00
AU Optronics (2409.TW)	E (02/10/2026)	NT\$23.15
Bizlink (3665.TW)	O (03/10/2025)	NT\$2,160.00
BOE Technology (000725.SZ)	O (09/06/2019)	Rmb5.83
Chenbro (8210.TW)	O (07/23/2025)	NT\$1,440.00
Chroma Ate Inc. (2360.TW)	O (10/05/2021)	NT\$2,190.00
E Ink Holdings Inc. (8069.TWO)	O (05/11/2026)	NT\$190.50
Ennostar Inc (3714.TW)	U (09/23/2022)	NT\$63.50
Hiwin Technologies Corp. (2049.TW)	O (03/30/2026)	NT\$318.50
Innolux (3481.TW)	E (04/07/2025)	NT\$46.95
King Slide Works Co. Ltd. (2059.TW)	O (11/08/2023)	NT\$6,400.00
Lens Technology (300433.SZ)	E (07/22/2020)	Rmb42.68
Radiant Opto-Electronics Corporation (6176.TW)	E (03/01/2024)	NT\$91.70
Sanan Optoelectronics (600703.SS)	U (08/21/2023)	Rmb15.26
TCL Corp. (000100.SZ)	E (04/07/2025)	Rmb4.51
Tianma Microelectronics (000050.SZ)	U (01/24/2018)	Rmb7.66
Wuhan Jingce Electronic Group Co Ltd (300567.SZ)	E (11/26/2021)	Rmb192.90
Howard Kao		
Acer Inc. (2353.TW)	U (04/23/2025)	NT\$36.90
Asustek Computer Inc. (2357.TW)	U (11/16/2025)	NT\$784.00
Compal Electronics (2324.TW)	U (04/23/2025)	NT\$36.15
FIT Hon Teng Ltd (6088.HK)	O (11/03/2025)	HK\$7.30
Giga-Byte Technology Co. Ltd. (2376.TW)	E (11/16/2025)	NT\$340.00
Gold Circuit Electronics Ltd. (2368.TW)	O (10/06/2022)	NT\$1,345.00
Inspur Electronic Information (000977.SZ)	E (08/28/2023)	Rmb57.84
Lenovo (0992.HK)	E (11/16/2025)	HK\$22.88
Lotes Co. Ltd. (3533.TW)	E (05/12/2025)	NT\$2,145.00
Nan Ya PCB (8046.TW)	O (02/23/2026)	NT\$807.00
Pegatron Corporation (4938.TW)	E (03/25/2026)	NT\$95.40
Quanta Computer Inc. (2382.TW)	O (05/01/2023)	NT\$370.00
Shengyi Technology Co Ltd. (600183.SS)	E (05/26/2022)	Rmb149.73
Shennan Circuits Co Ltd (002916.SZ)	E (08/24/2023)	Rmb376.16
Unimicron (3037.TW)	O (02/23/2026)	NT\$854.00
Wistron Corporation (3231.TW)	O (07/12/2023)	NT\$152.50
Wiwynn Corp (6669.TW)	O (11/10/2025)	NT\$4,900.00
Yageo Corp. (2327.TW)	O (10/28/2025)	NT\$842.00
Zhen Ding (4958.TW)	O (05/18/2026)	NT\$529.00
Sharon Shih		
Asia Vital Components Co. Ltd. (3017.TW)	O (07/30/2024)	NT\$2,335.00
Auras Technology Co Ltd (3324.TWO)	E (05/04/2023)	NT\$998.00
Catcher Technology (2474.TW)	U (11/17/2025)	NT\$205.50
Delta Electronics Inc. (2308.TW)	O (07/13/2017)	NT\$2,160.00
Fositek Corp (6805.TW)	O (06/25/2025)	NT\$1,750.00
Foxconn Industrial Internet Co. Ltd. (601138.SS)	O (07/10/2019)	Rmb69.52
Foxconn Technology (2354.TW)	U (04/23/2025)	NT\$56.20
GoerTek Inc (002241.SZ)	U (04/23/2025)	Rmb22.72
Hon Hai Precision (2317.TW)	O (03/15/2024)	NT\$258.50
LandMark Optoelectronics Corporation (3081.TWO)	E (03/26/2026)	NT\$2,165.00
Lingyi Itech Guangdong Co (002600.SZ)	U (04/23/2025)	Rmb13.63
Lite-On Technology (2301.TW)	E (01/15/2025)	NT\$208.50
Luxshare Precision Industry Co., Ltd. (002475.SZ)	O (10/24/2016)	Rmb64.58

Sunonwealth Electric Machine Industry Co (2421.TW)	E (02/23/2024)	NT\$142.00
Tong Hsing (6271.TW)	E (03/18/2019)	NT\$217.00
Visual Photonics Epitaxy Co Ltd (2455.TW)	E (09/11/2023)	NT\$320.00

Stock Ratings are subject to change. Please see latest research for each company.
* Historical prices are not split adjusted.

© 2026 Morgan Stanley